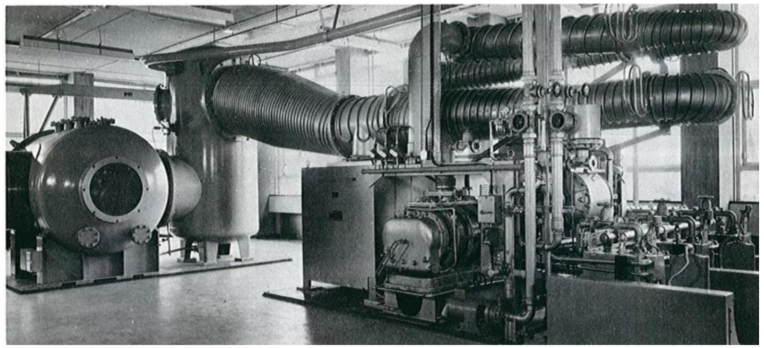
Everything old is new again – Recreating UK Hypersonic Rarefied Capability

A/Prof Luke J Doherty

Incl. contributions from Dr Nathan Donaldson (FGE), Alec Berry (4YP), Tracey Saber (4YP), Nnaemeka Anyamele (DPhil), Prof Peter Ireland and many older Oxford DPhils...

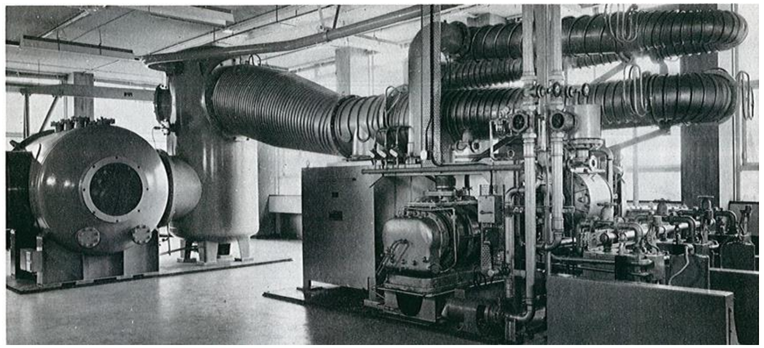
UK Rarefied Facilities – Then and now

UK Rarefied Facilities – **Then** and now



Oxford (1963)

UK Rarefied Facilities – **Then** and now



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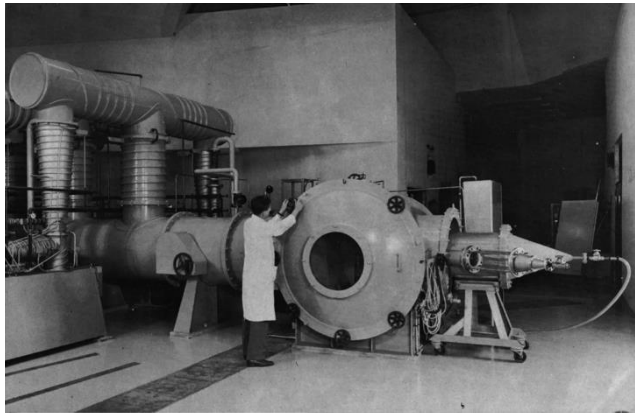

NPL (~1965)

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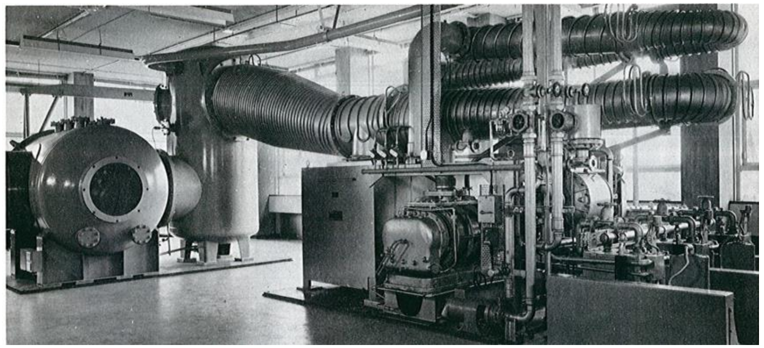

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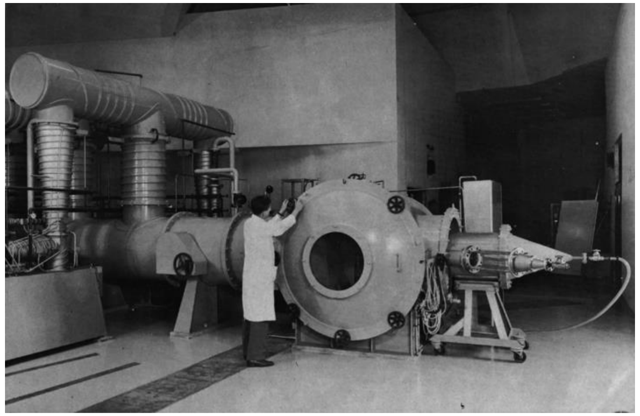

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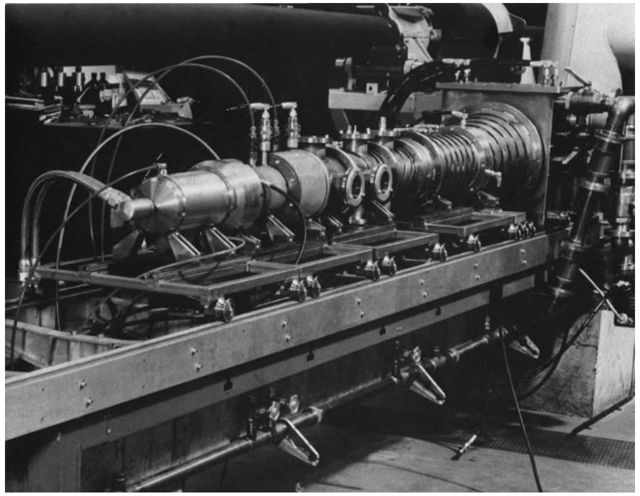
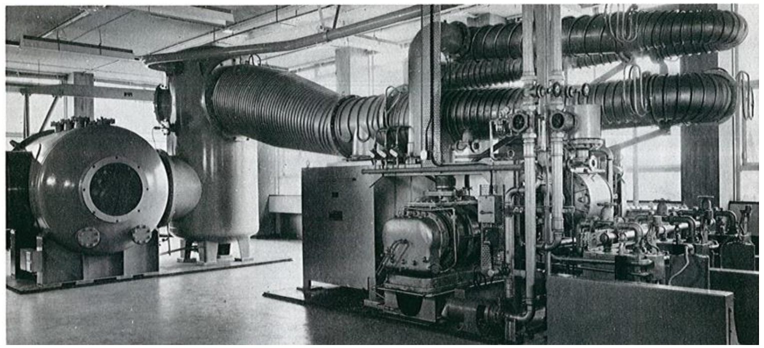


Imperial College London (~1972)

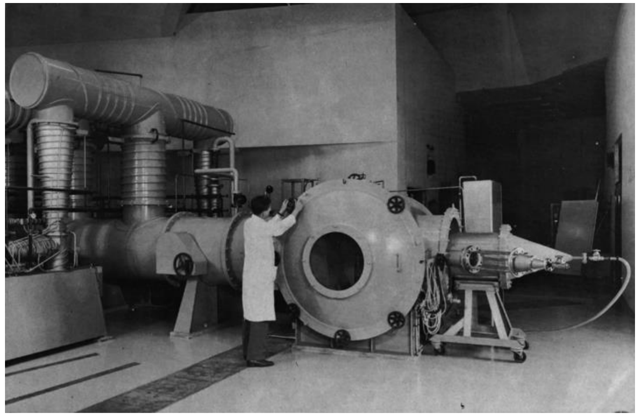
UK Rarefied Facilities – **Then** and now



Oxford (1963)

?

RAE (19??)



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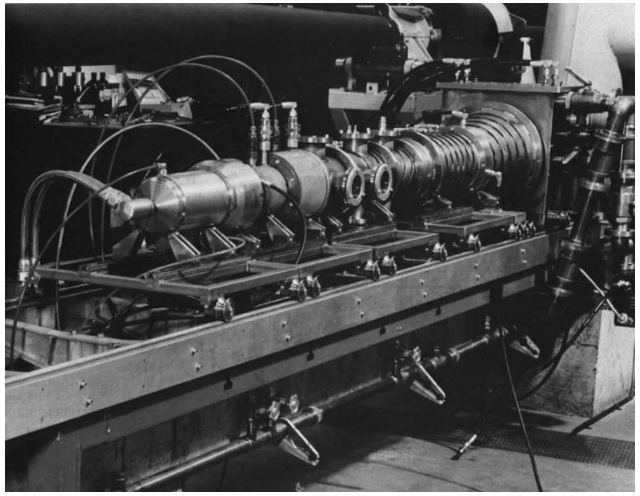


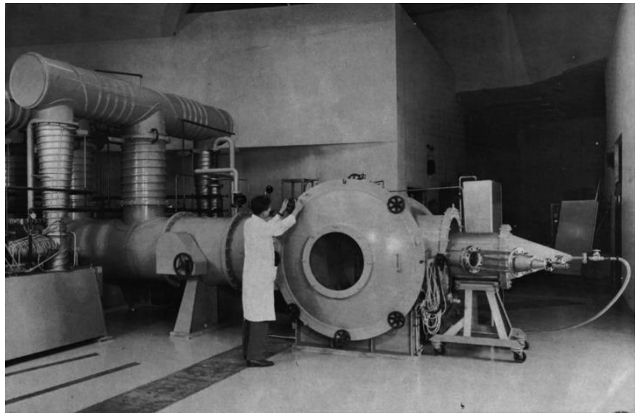
Imperial College London (~1972)

UK Rarefied Facilities – Then and **now**

Oxford (1963)

?

RAE (19??)



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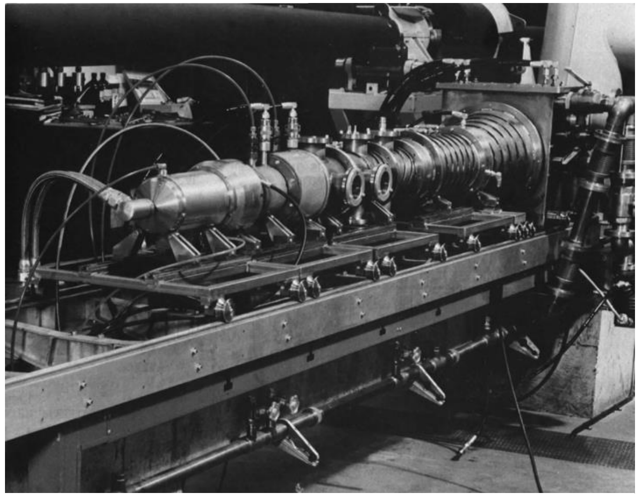
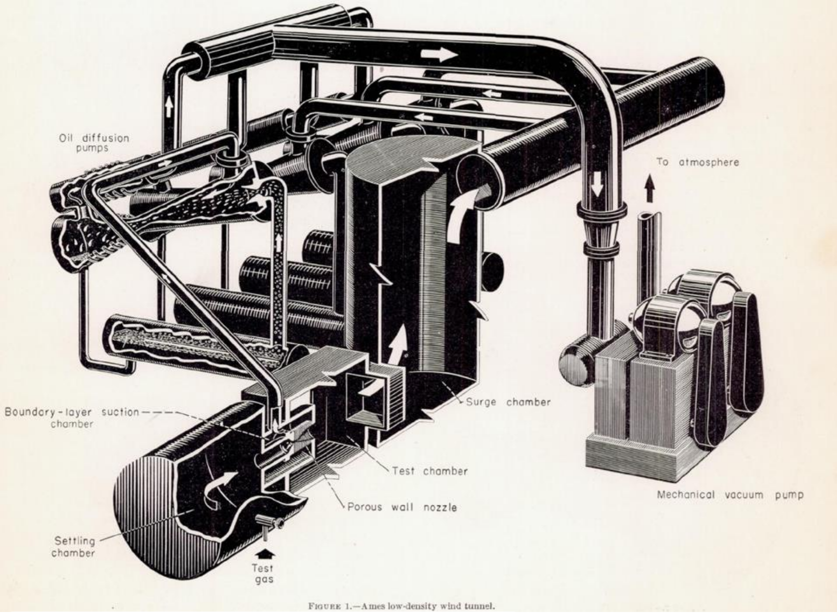


Imperial College London (~1972)

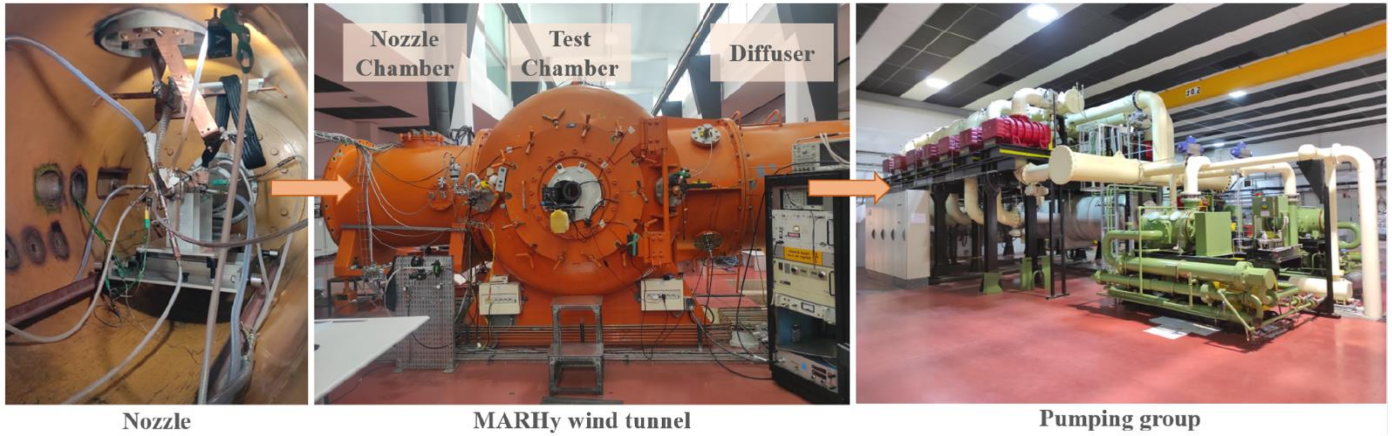
International Facilities – **Then** and Now

DEPARTMENT OF ENGINEERING SCIENCE
UNIVERSITY OF OXFORD



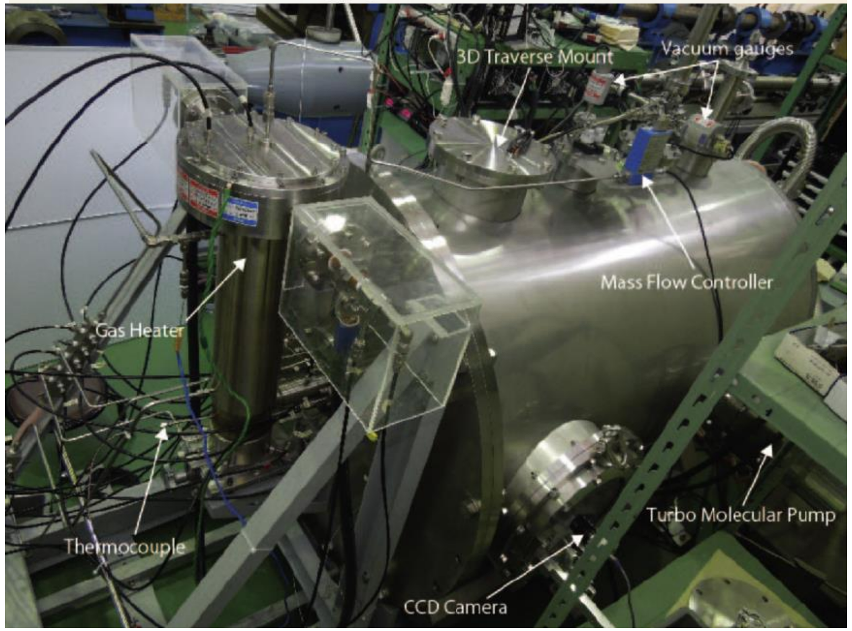
USA: NASA AMES (1952)

Germany: DLR VxG (1964 - 1970)



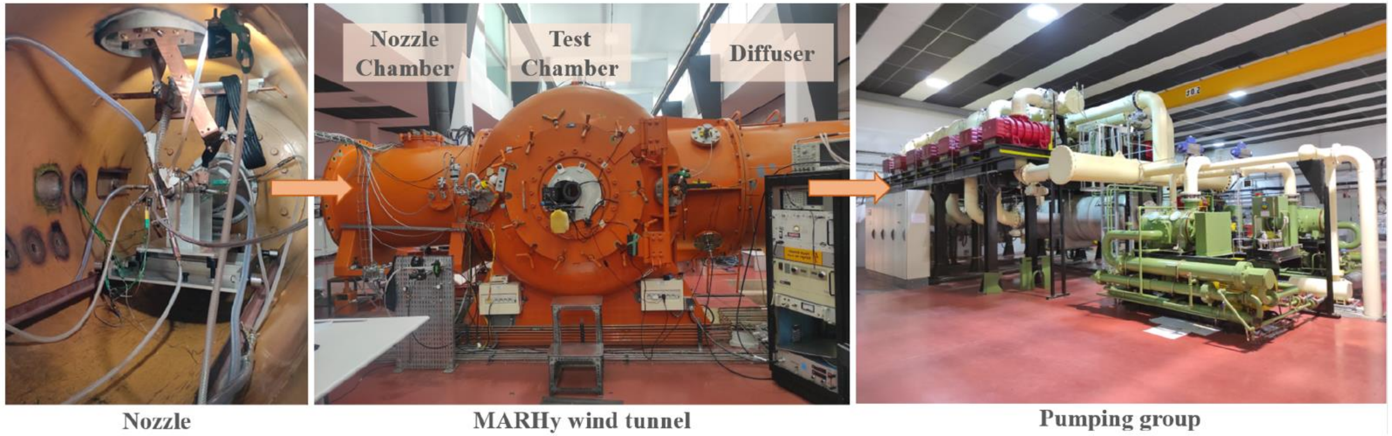
France: SR3/MARHy (1963)

International Facilities – Then and **Now**



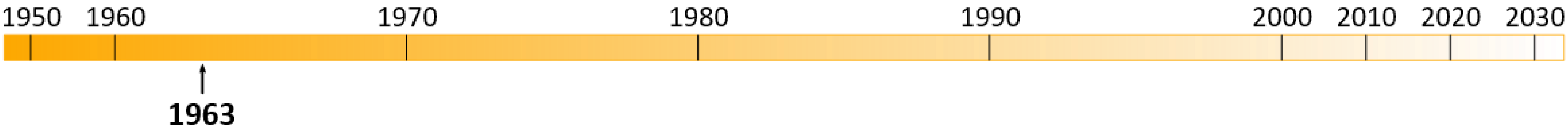
Japan: JAXA HRWT (~2015)

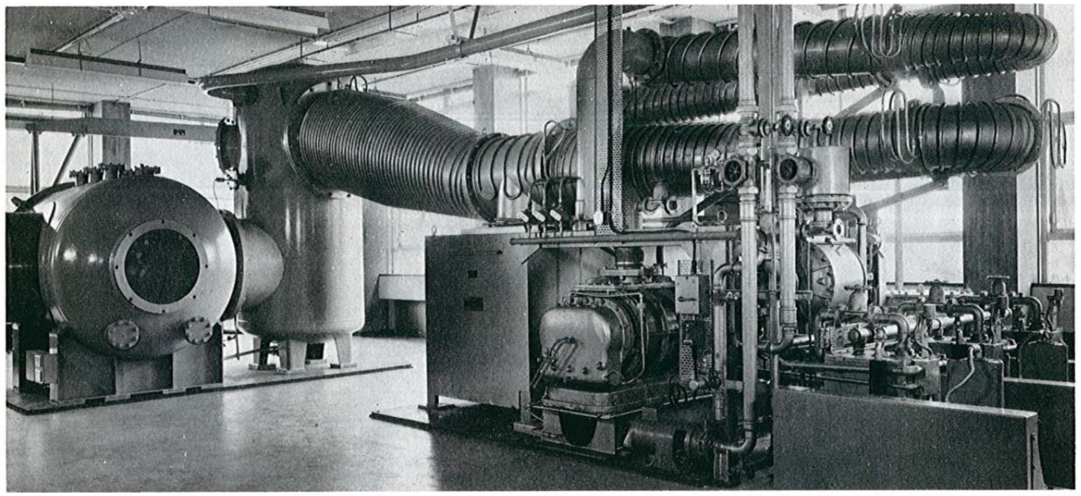
Germany: DLR VxG (1964 - 1970)



France: SR3/MARHy (1963)

Oxford Low Density Tunnel (LDT)



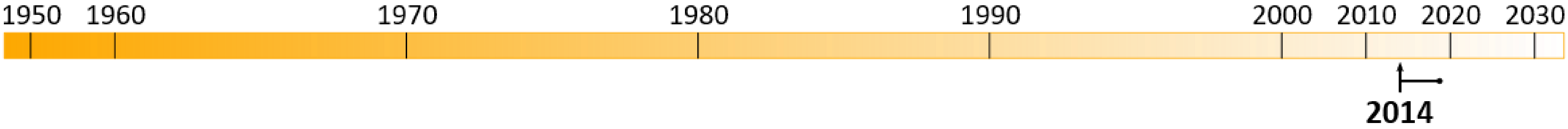


Edwards Model 100B4 vapour booster installed at Oxford University with the top body cone swung through 90° to clear low headroom.

Photograph of LDT installed at Oxford circa 1967

- Delivered to Department of Engineering Science in 1963
- Continuous, free-jet, open circuit facility
- Three stage vacuum pumping system:
  - Vapour booster + Roots + 3 x piston rotary backing pumps
  - 22,000 L/s speed, w_{max} ~ 0.6 g/s (air)
- Operates in hypersonic rarefied slip and transition regime (cold-flow)
  - Free-molecular flow conditions possible

Oxford Low Density Tunnel (LDT)

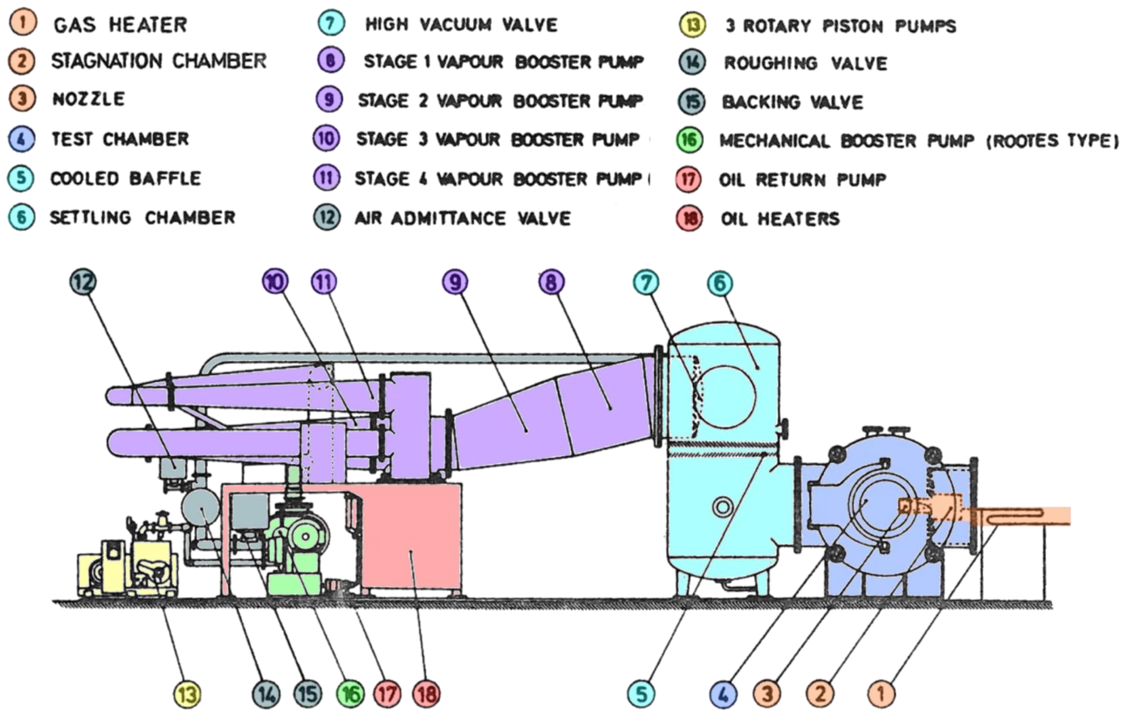


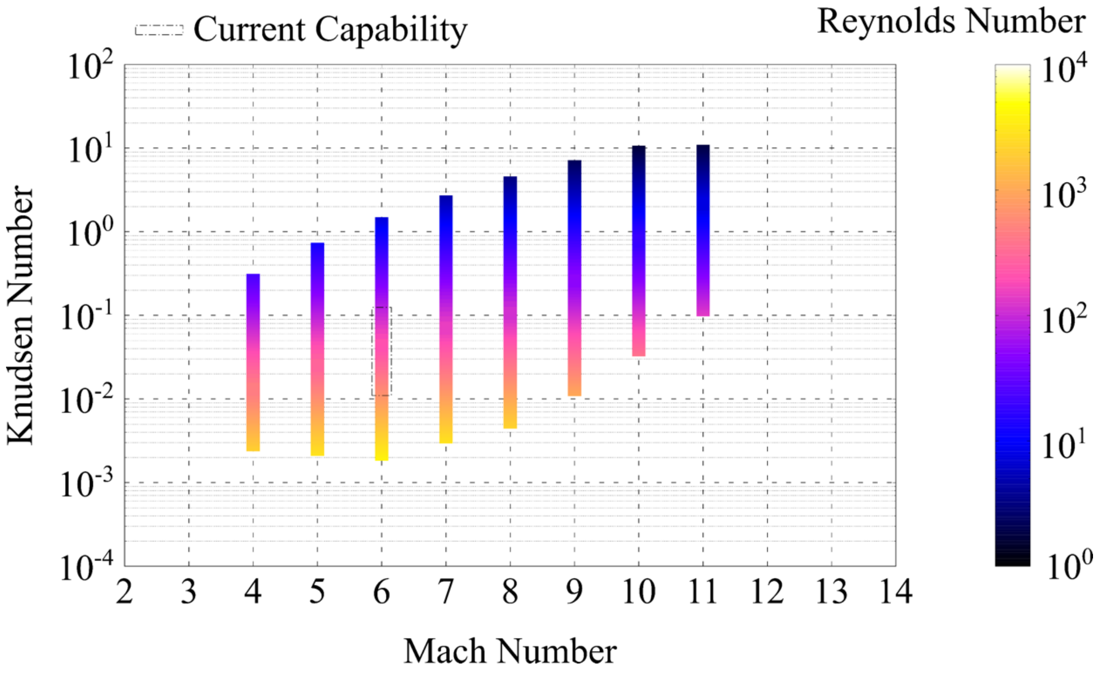
Photograph of LDT as currently installed at Oxford

NWTF

- 2014 UKRI/EPSRC National Wind Tunnel Facility (NWTF) grant awarded
- Oxford built 2x ("new") high speed wind tunnel facilities –
  - High Density Tunnel (HDT)
  - T6 Stalker Tunnel
- LDT refurbished and recommissioned by Dr Nathan Donaldson during his DPhil –
  - DAQ, instrumentation, gas inlet heater, control system, vapour booster oil (£££), seals, cooling water, asbestos insulation, ...

Oxford Low Density Tunnel (LDT)





Theoretical performance map of LDT for L_{ref} = 10 mm (with a few assumptions...)

A whistle stop tour of Research

(conducted in LDT)

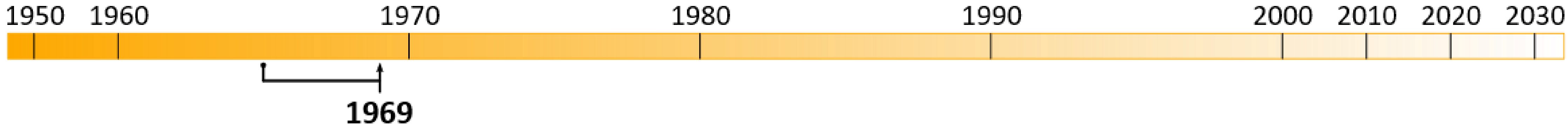
- G.P.D. Rajasooria
- Magnetic Suspension & Balance System
  - H. Altman
  - K.S. Hadjimichalis
  - T.F. Haslam-Jones
  - G. Dahlen
  - A. Owen

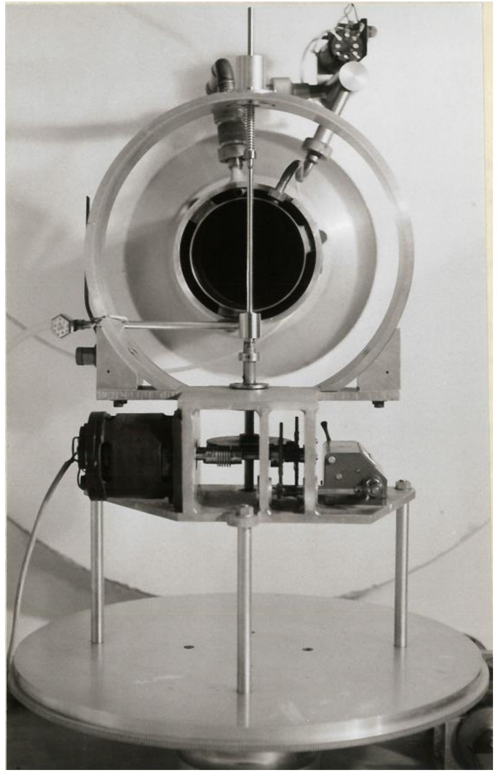
Original Era
(1963 – 2003-ish)

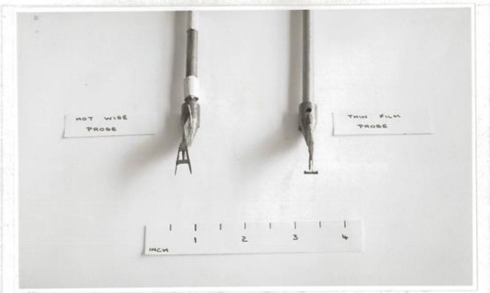
- Nathan Donaldson
- Alec Berry (4YP)
- Tracey Saber (4YP)
- Nnaemeka Anyamele

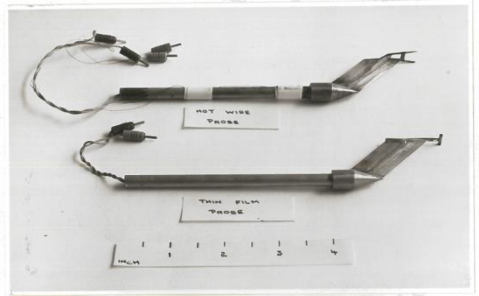
NWTF Era
(2014 – now)

G.P.D. Rajasooria - Wakes









- Surface pressure, impact (Pitot) pressure, thin-film, & hot-wire measurement surveys
- Derived velocity, pressure, and temperature profiles
- Kn = 0.0112 or 0.0056

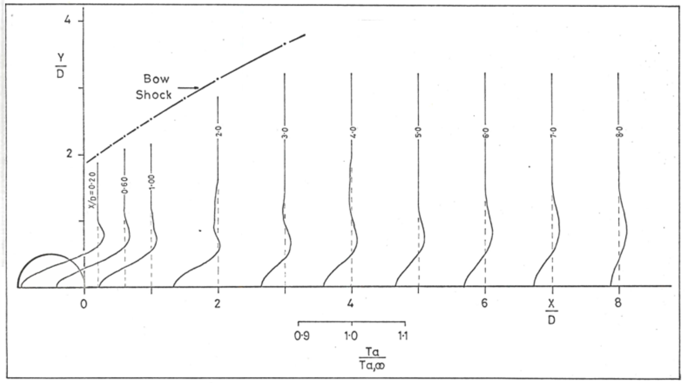


Fig. 5-16. Thin film probe adiabatic temperature profiles (0.250 inch cylinder, $Re_{\infty,D}$ = 758).

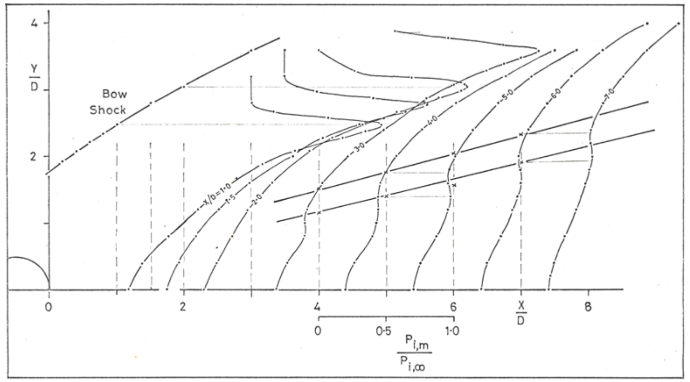
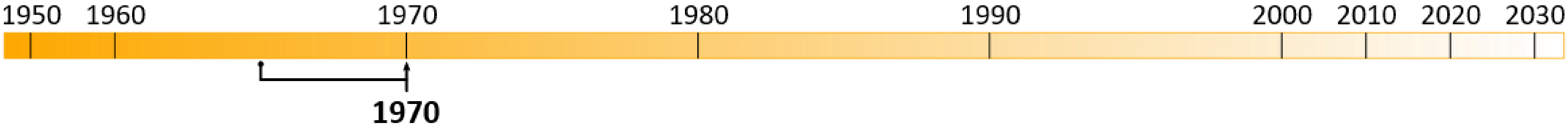
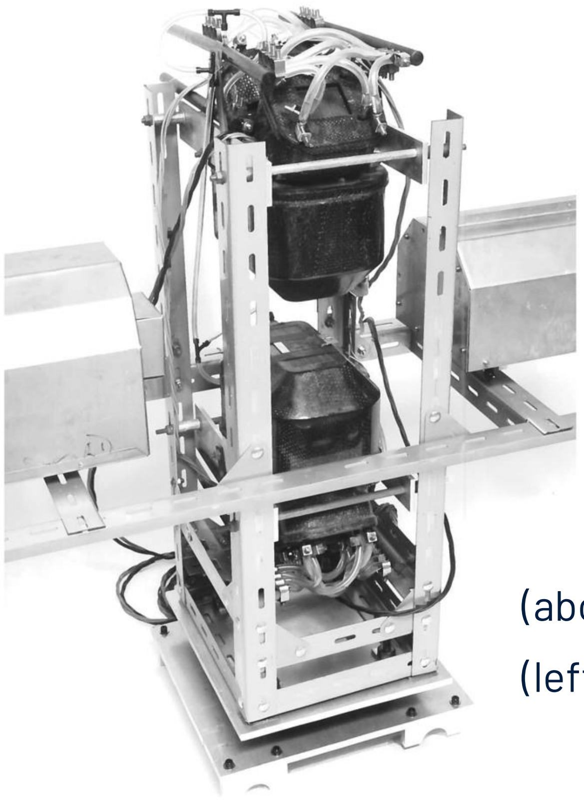
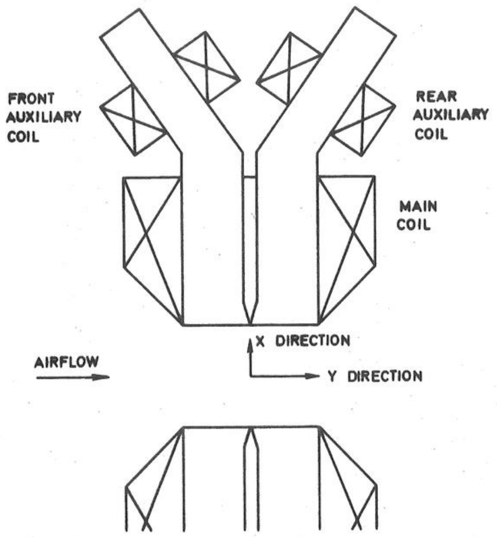


Fig. 5-14. Pitot pressure profiles (0.250 inch cylinder, $Re_{\infty,D}$ = 758).

H. Altmann – Sphere Drag using an MSBS



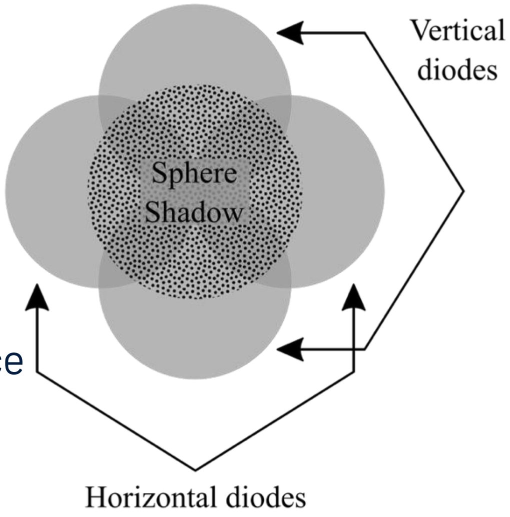




- Designed Oxford's 1st Magnetic suspension and balance system (MSBS) between 1965 – 1970, further development until 1973
- Spheres only

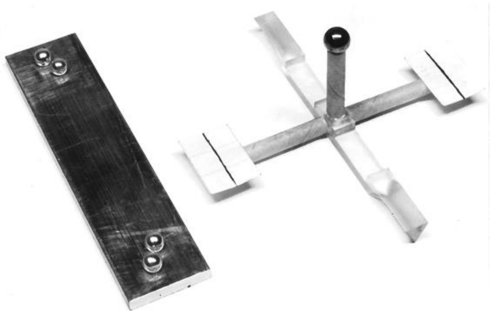
(above) Magnet layout

(left) Photograph of complete balance

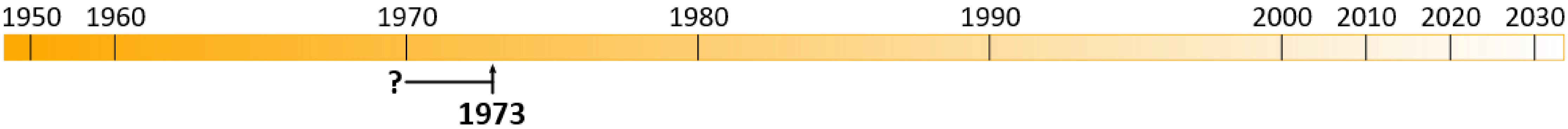


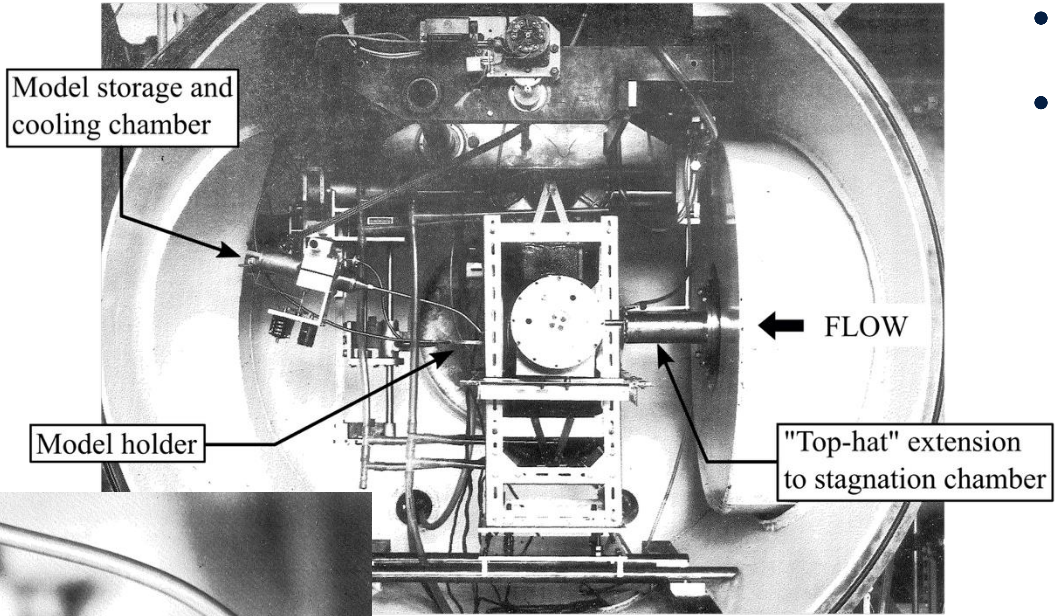
(left) Model detection system

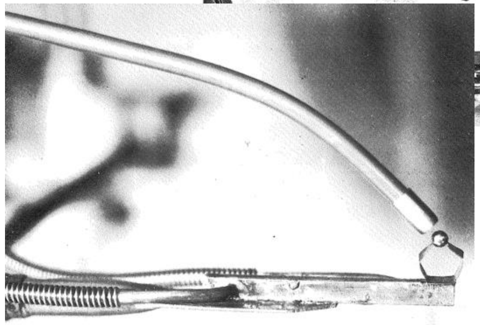
(below) Calibration balance



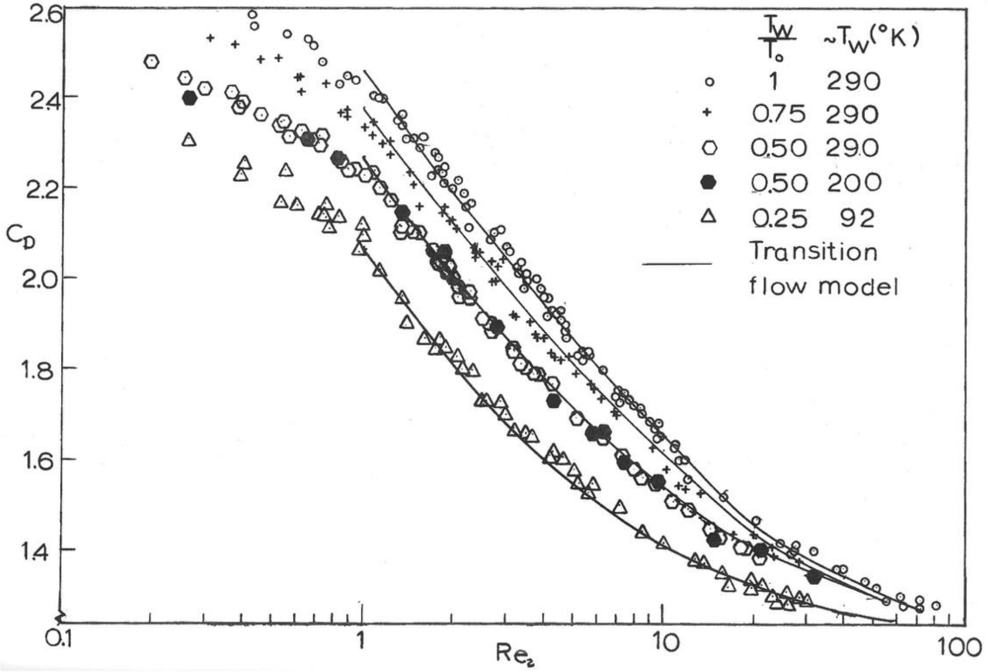
K.S. Hadjimichalis – Sphere Drag (#2)







- Extensive testing utilising the 1st-gen MSBS
- Continuum → Free-molecular; T0/Tw 0.25 → 1



(above) Set up in LDT test section

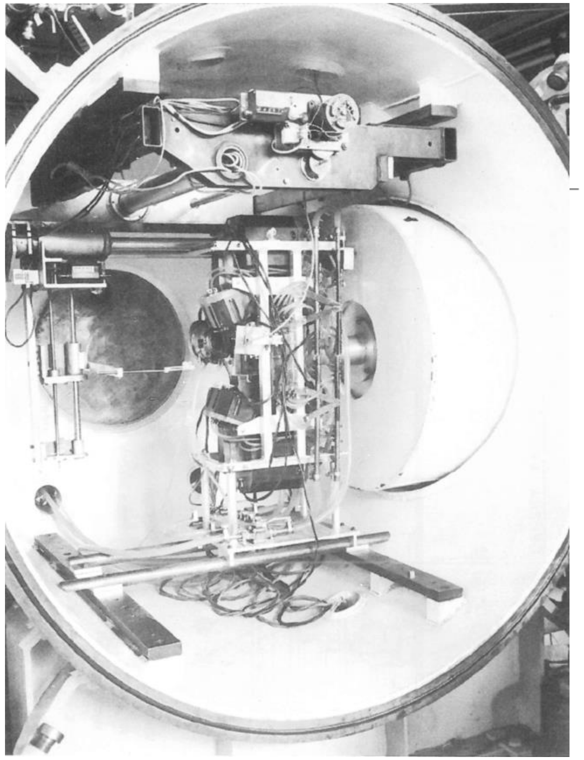
(left) LN2 cooled model holder

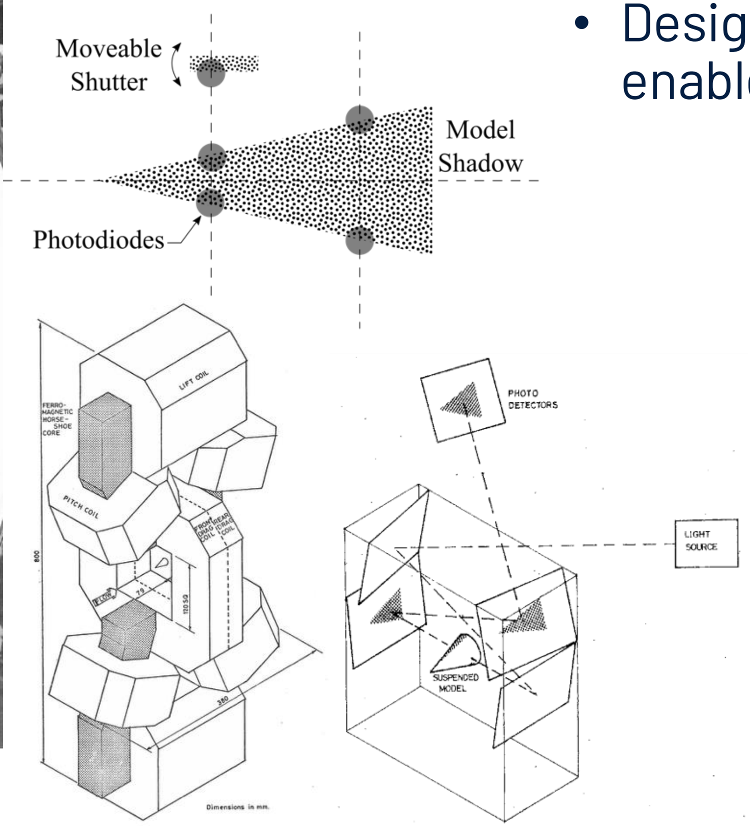
(right) Experimental results

T.F. Haslam-Jones – Slender Cone Drag

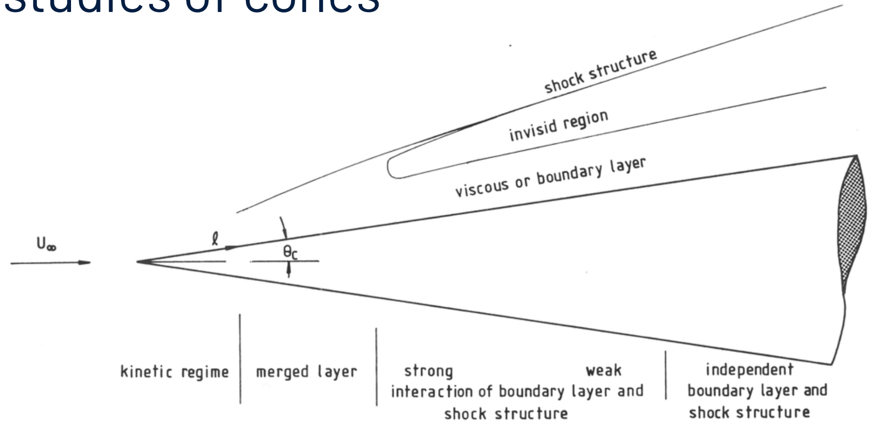
1950 1960 1970 1980 1990 2000 2010 2020 2030

1978





- Designed 2nd-gen MSBS (3-component) to enable studies of cones

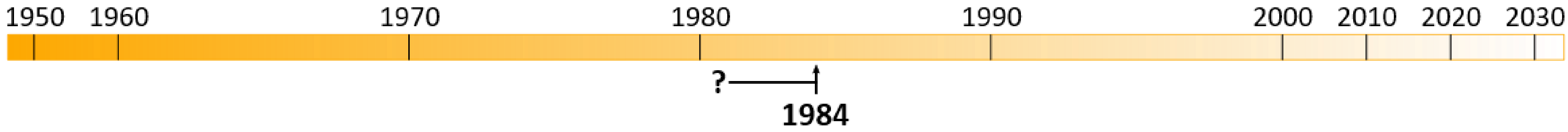


(far-left) MSBS installed in LDT test section

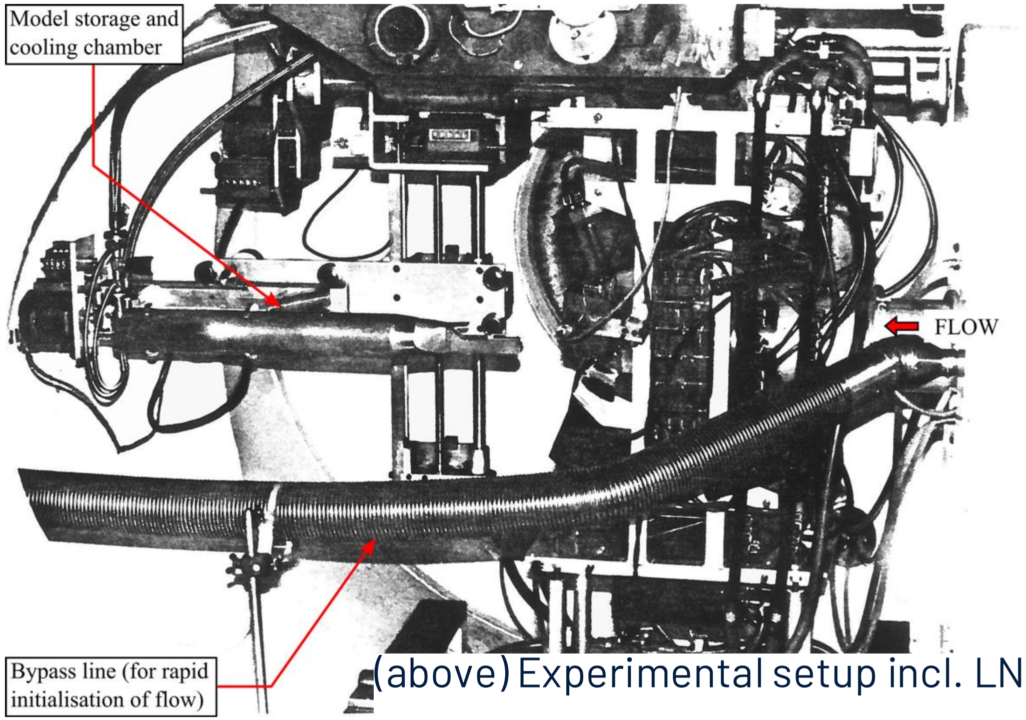
(left; upper and lower) Model position detection system and mirrors

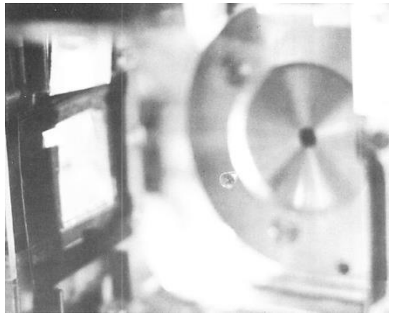
(above) Cone flow-field at rarefied conditions

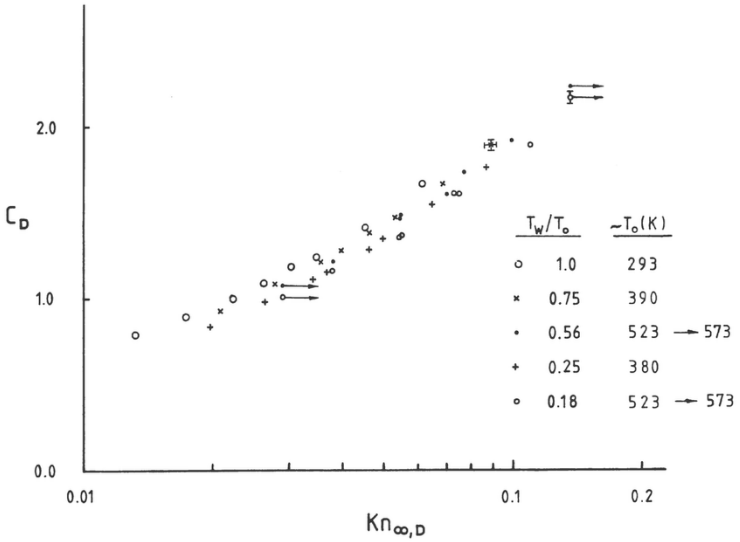
G. Dahlen – Cone Drag (#2)



- Undertook extensive testing investigating –
  - wall-to-total temperature ratio
  - nose bluntness





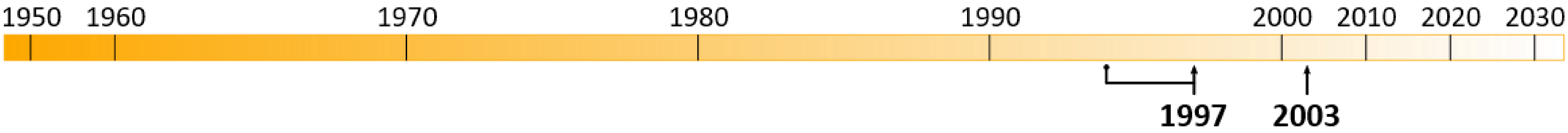


(above) Experimental setup incl. LN2 cooling chamber

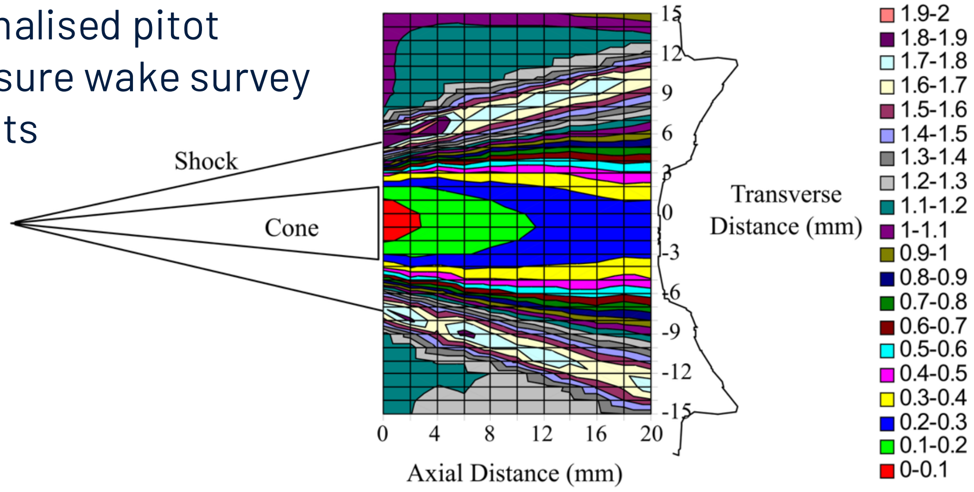
(right) Suspended cone

(far-right) Results for 3° sharp cone

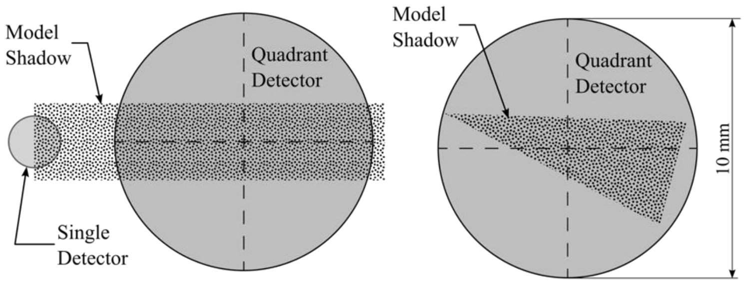
A. Owen – Re-entry Vehicle Aerodynamics



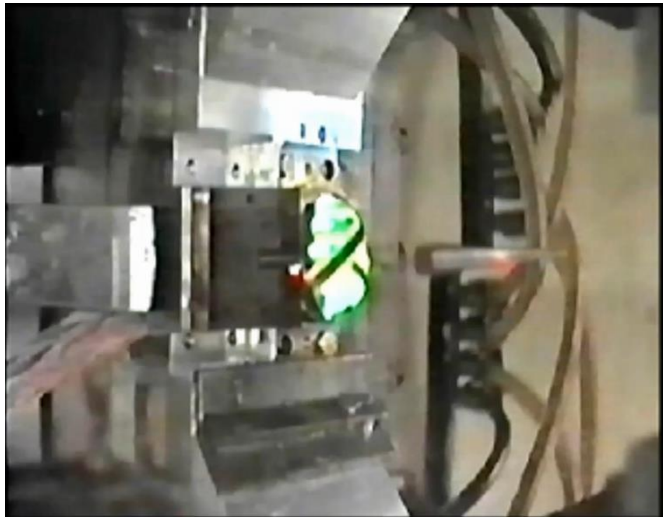
Normalised pitot pressure wake survey results

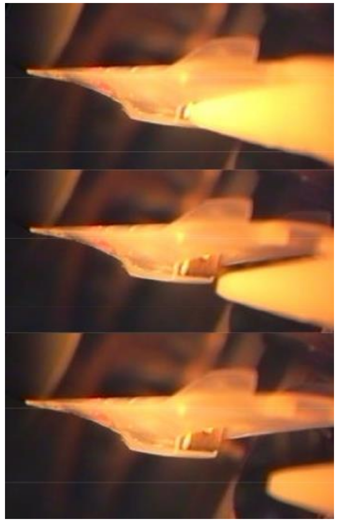


- Recommissioned the MSBS
- Refined model position detection system
- Extended the capability to non-conical geometries
  - Cones at AoA; Aerobrake; X-43A (2% scale)

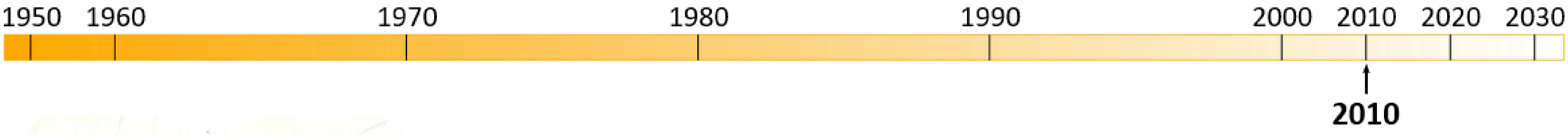


Updated model detection system



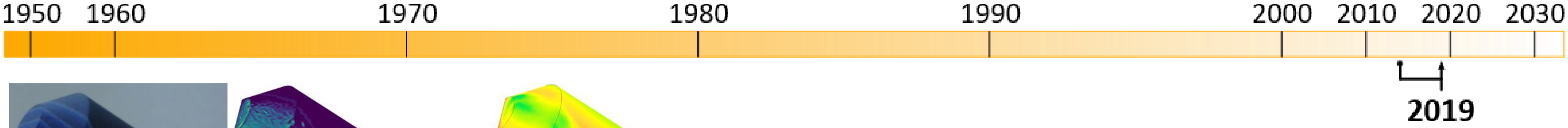


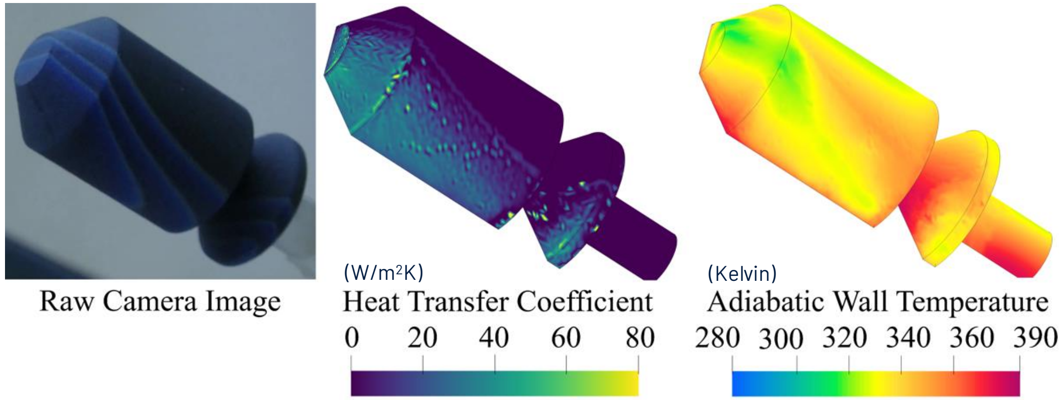
“Interlude”



- The Oxford Thermofluids Institute (OTI) aka Osney Lab moves from Old Power Station premises into current location
  - LDT and MSBS (3-DoF) were kept in the transition
  - Group had been based in the Power station since circa-1974

Nathan D. – Satellite Heat Transfer





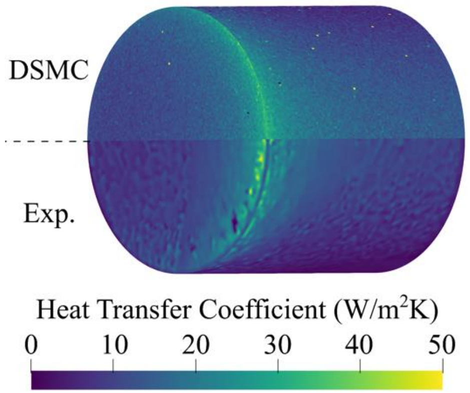
- Recommissioned LDT after a decade of inactivity
- Established use of Liquid Crystal Thermography (LCT) in rarefied flow (single-camera)
- Extensive nozzle characterisation

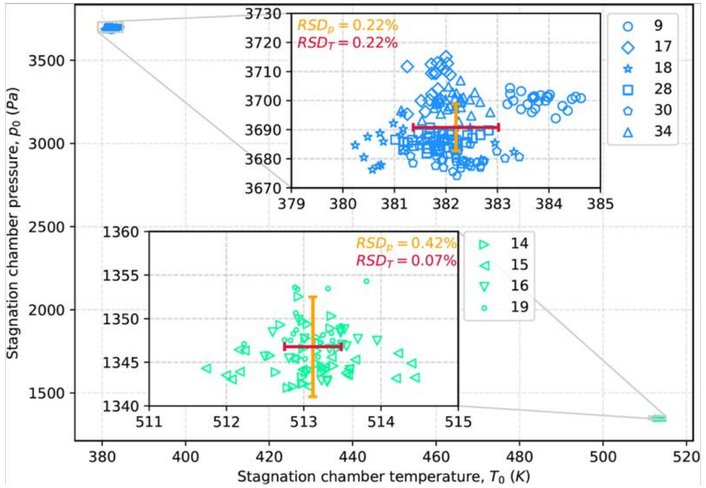
(above) Upper stage

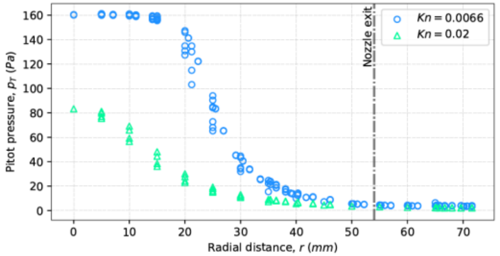
Kn = 0.02, AoA = 60°

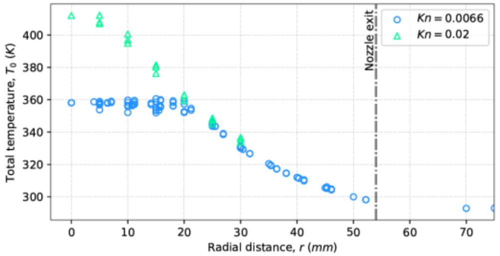
(left) Flat faced Cylinder

Kn = 0.1, AoA = 45°









Science and Technology Facilities Council

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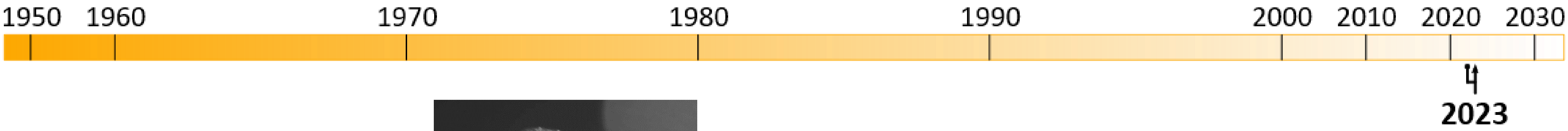
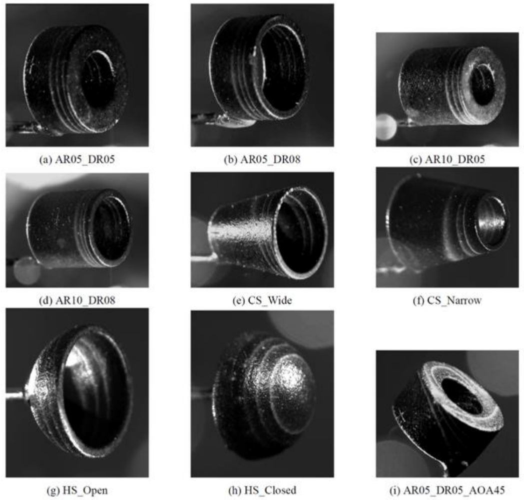
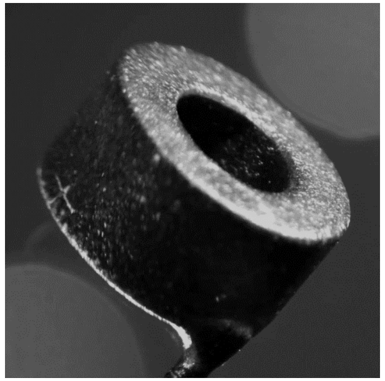
Alec B. (4YP) – Satellite Heat Transfer (#2)

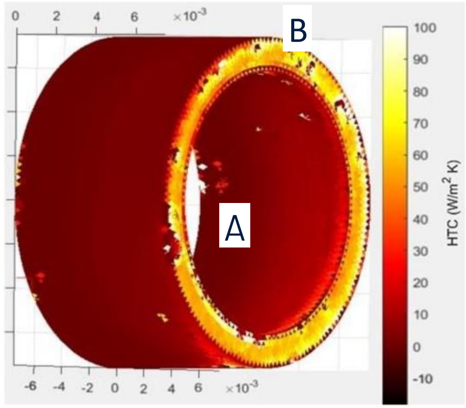


- Extended LCT method to multiple cameras
- Tested a range of hollow geometries

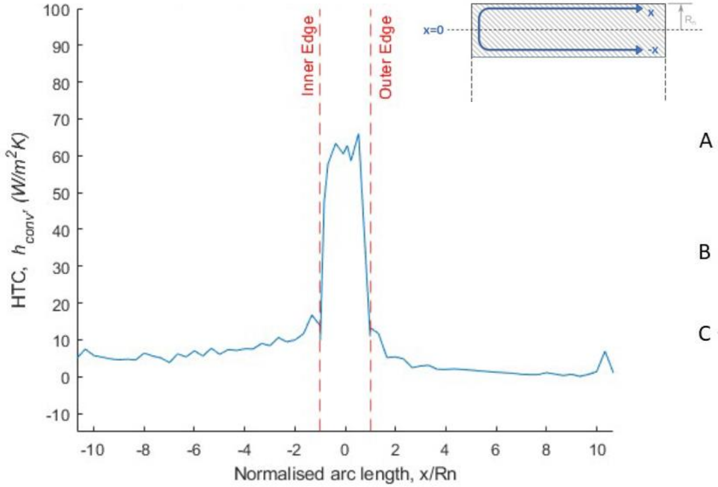


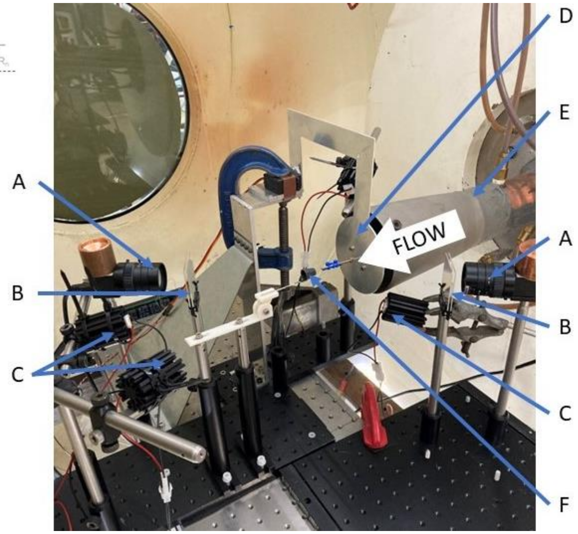


Transition temps
30°C, 35°C, 40°C



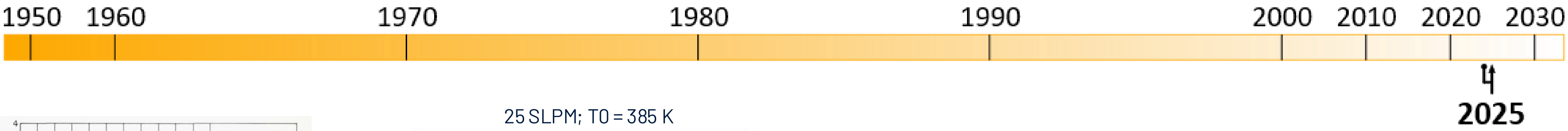
Length ratio = 0.5
Diameter ratio = 0.8
(A) Numerical anomalies
(B) Discontinuities

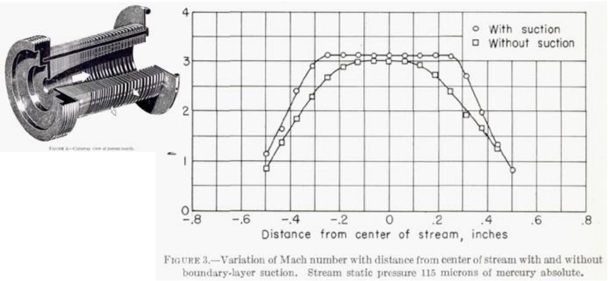




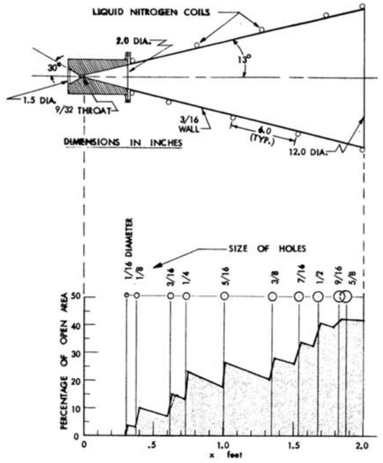
A – camera; B – optical filter; C – LED;
D – Flow Shield; E – Nozzle; F - Model

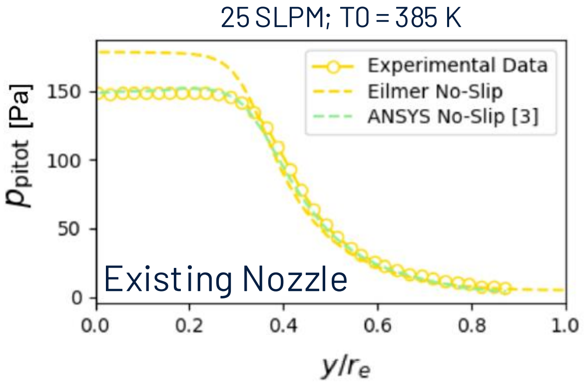
Tracey S. (4YP) – Nozzle Design (w/bleed)

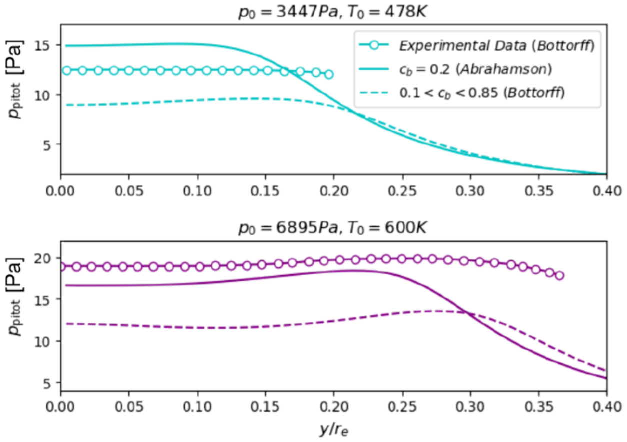




(Above) Stalder et al;

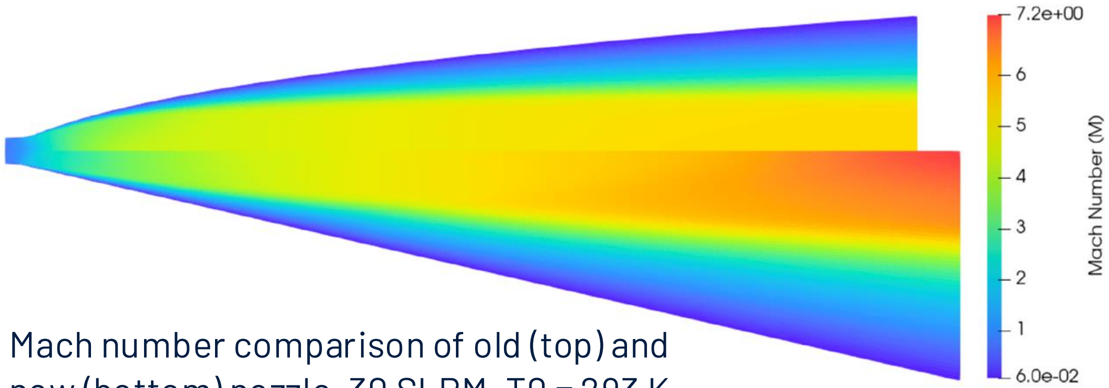






Bottorff and Rogers nozzle (left) and Eilmer "validation" (right)

- Goal: a new nozzle with improved core flow size
- Inspired by Stalder et al (1952) and Bottorff & Rogers (1964)
- Utilised Uni. of Qld Eilmer code with UDF BC
- Lots of challenges...



Mach number comparison of old (top) and new (bottom) nozzle. 30 SLPM, T0 = 293 K

Nnaemeka A. – Swept SBLI

DEPARTMENT OF ENGINEERING SCIENCE
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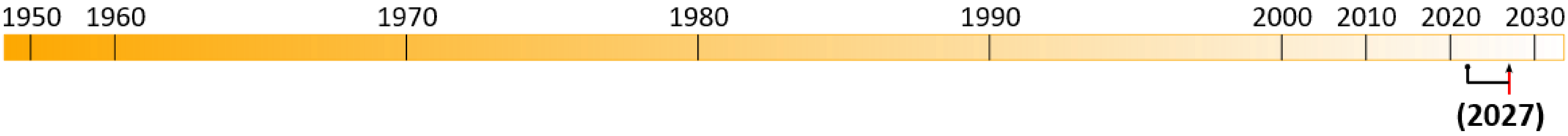
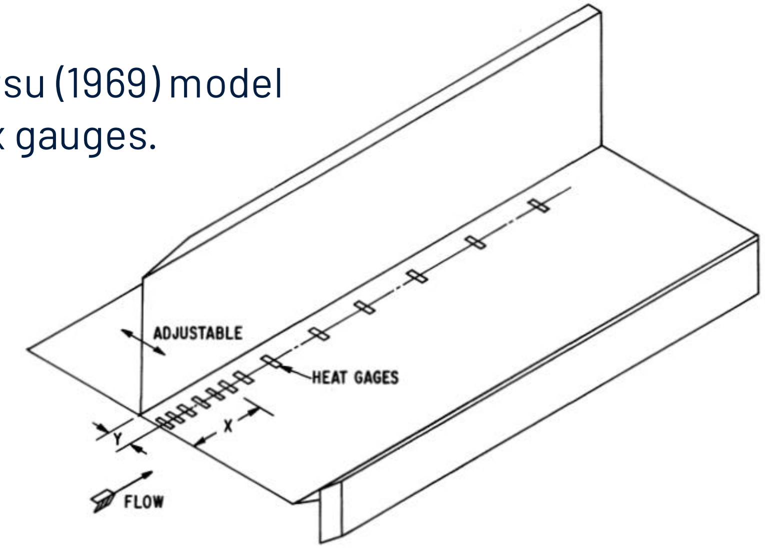


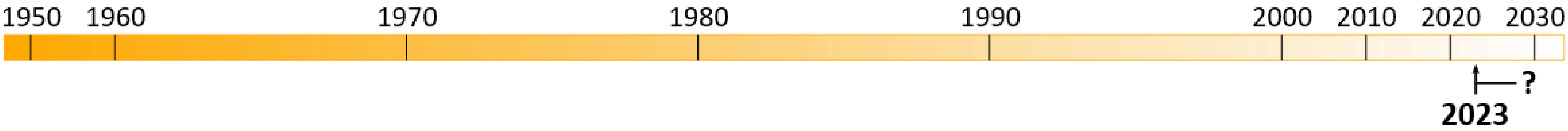
(Right) Nagamatsu (1969) model with Pt heat flux gauges.

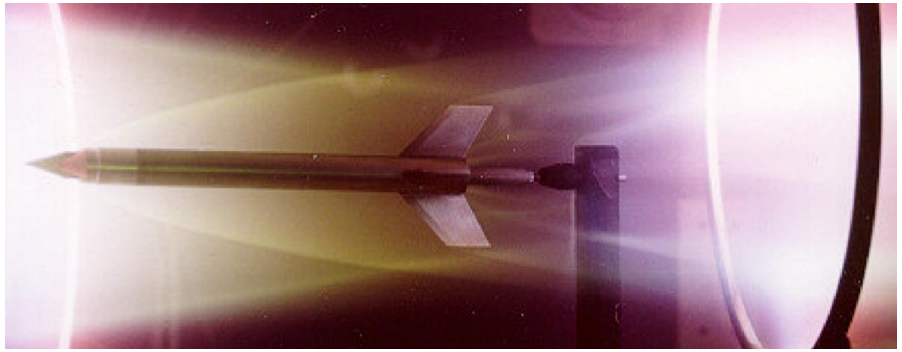


LDT results coming soon!

- Goal: to understand Swept SBLI in rarefied flows
- Experimental data largely absent from literature
- Exceptions: Nagamatsu (1969) and Schmidt (1971)
- No computational studies in public literature
- Planning for –
    - Pitot survey
    - Hot-wire surveys
    - Surface heat transfer (via LCT)
- Completed various facility repairs and upgrades (DAQ software, processing, documentation etc)

“Recent” events & Future outlook



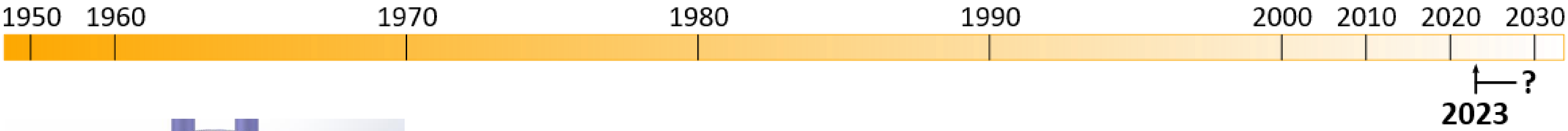


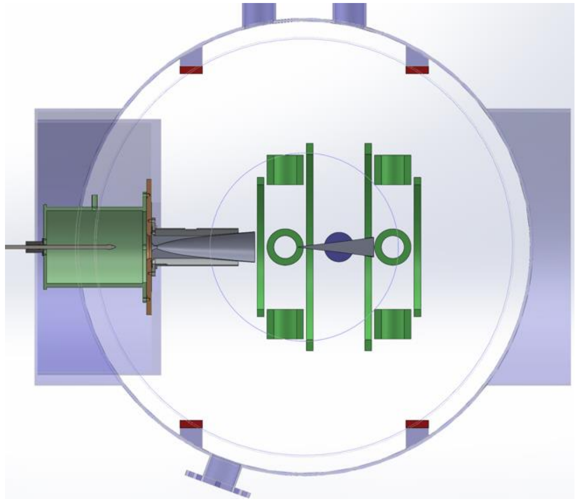
GDF image of HyShot vehicle stack in QinetiQ LDT (Cain et al 2005)

Royal Society – Equipment Funding

- Cameras for LCT
    - 4x Kaya Iron250M, 5 MP, 133 fps (10 bit)
    - 2x Kaya Iron4521M, 21 MP, 186 fps (10 bit)
    - Dedicated recording system (hardware on-site)
- Glow Discharge Fluorescence (GDF)
    - -6kV, 500mA HV power supply (scary! PO issued)
    - Electrodes, safety interlock are a work in progress
    - Open to suggestions for geometries to test…

"Recent" events & Future outlook





(Above) Concept 6-DoF MSBS in LDT Test Section

(Right) Prototype porous conical nozzle

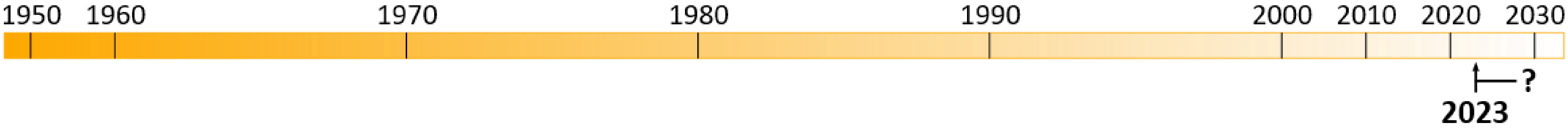
NWTF+ MSBS

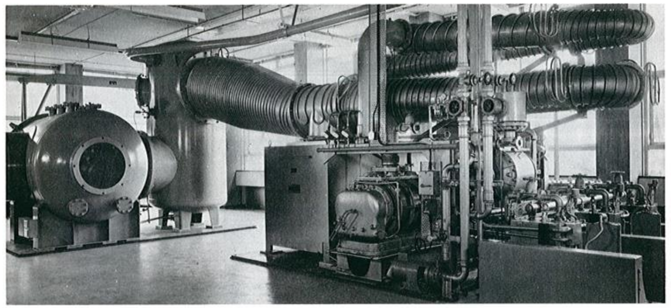
- ~£23M investment in both new and existing facilities
- 2 x MSBS included → 1x Imperial + 1x Oxford

Multi-engine Rocket Plume Interactions

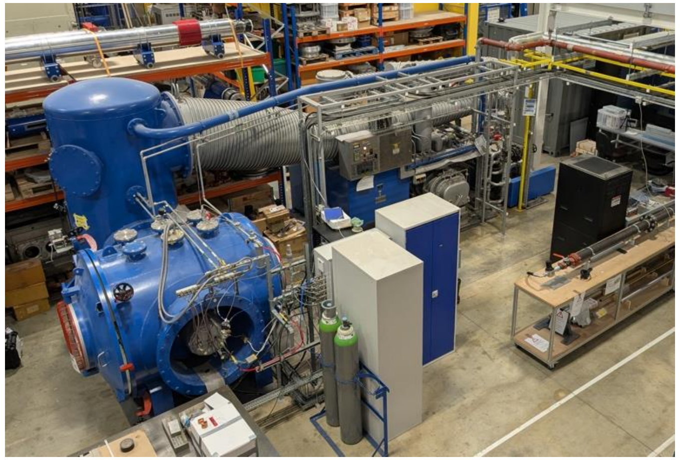
- Tracey Saber DPhil, incoming Clarendon Scholar
- Collaboration with DLR
- Combined experimental/numerical study utilising hybrid Fokker-Planck/DSMC code

Final Thoughts…





Edwards Model 100B4 vapour booster installed at Oxford University with the top body cone swung through 90° to clear low headroom.



1. (Re) Building knowledge and capability is much **harder** than maintaining capability
2. Maintaining capability requires a **critical mass** of users and developers
3. The future of the facility can only be **secured** by increased usage
4. As an NWTF, the LDT is **open for business**

How can LDT help your research?

With thanks to...

DEPARTMENT OF ENGINEERING SCIENCE

UNIVERSITY OF OXFORD

Ministry of Defence

[dstl] Delivering Mission Success

UKRI Science and Technology Facilities Council

THE ROYAL SOCIETY

FLUID GRAVITY ENGINEERING LTD

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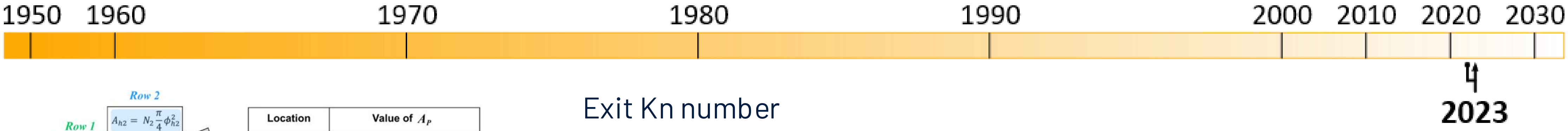
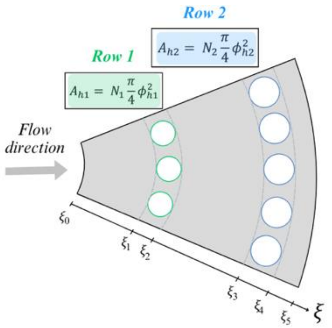
5) Schmidt, E.M., "Merged Flow in an Interior Corner," PhD Thesis, Polytechnic Institute of Brooklyn, 1969.

Extras

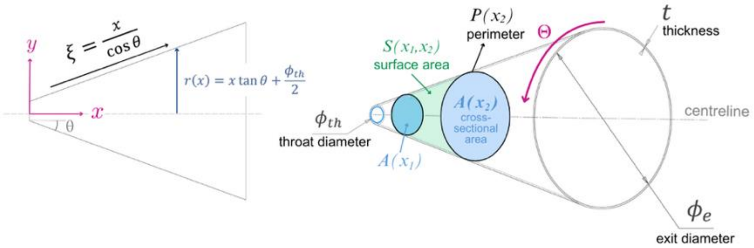
1. Tracey Saber – 4YP nozzle design
2. 3 DoF MSBS re-commissioning
3. RAE 6 DoF MSBS
4. Overview of Oxford Hypersonic Facilities

T. Saber (4YP) – Nozzle Design (w/ bleed)





Location	Value of A_P
x_0, ξ_0	0
x_1, ξ_1	0
x_2, ξ_2	$\frac{A_{h1}}{S(x_1,x_2)}$
x_3, ξ_3	$A_P(x_2)\frac{P(x_2)}{P(x_3)}$
x_4, ξ_4	$\frac{A_{h2}}{S(x_3,x_4)}$
x_5, ξ_5	$A_P(x_4)\frac{P(x_4)}{P(x_5)}$



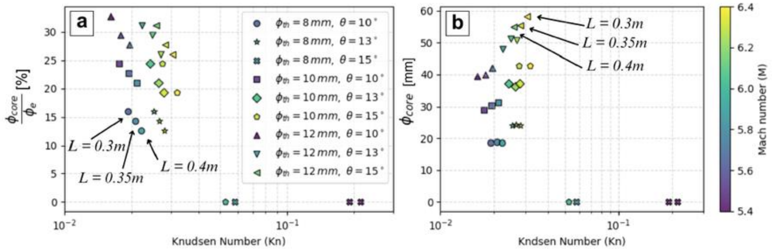


Figure 25: Comparison of several solid-wall nozzle geometries for $p_0 = 4000Pa,\ T_0 = 470K$

Exit Kn number

	$T_0 = 293K$	$T_0 = 470K$
Current Contoured Nozzle	0.008	0.014
New Optimised Perforated Nozzle	0.0197	0.033

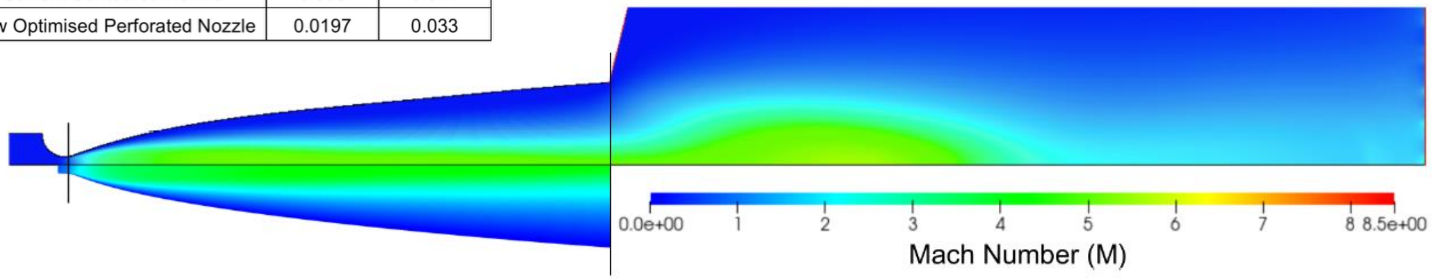
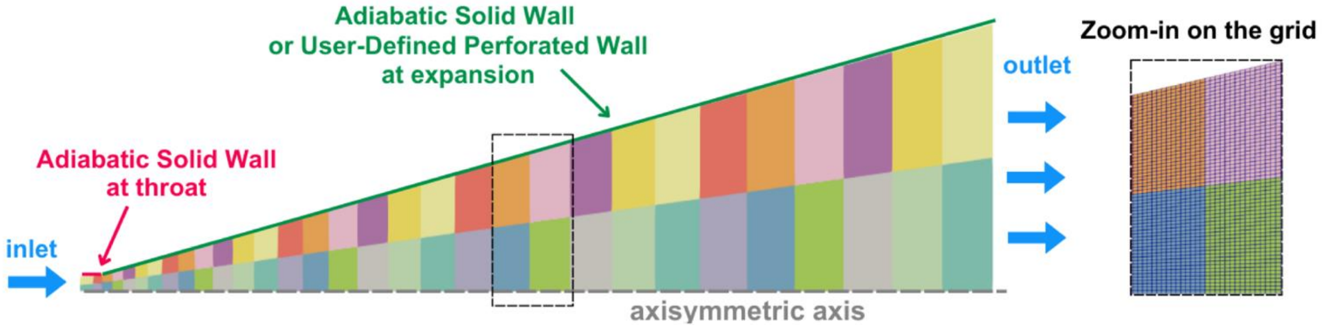
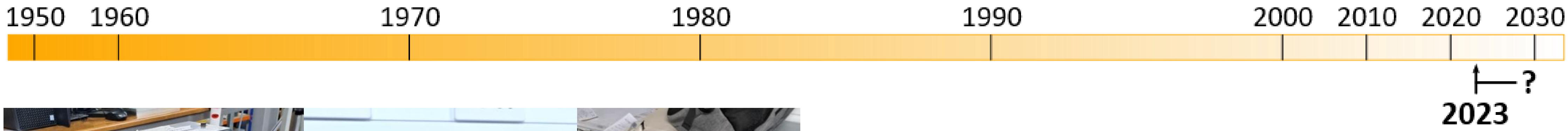
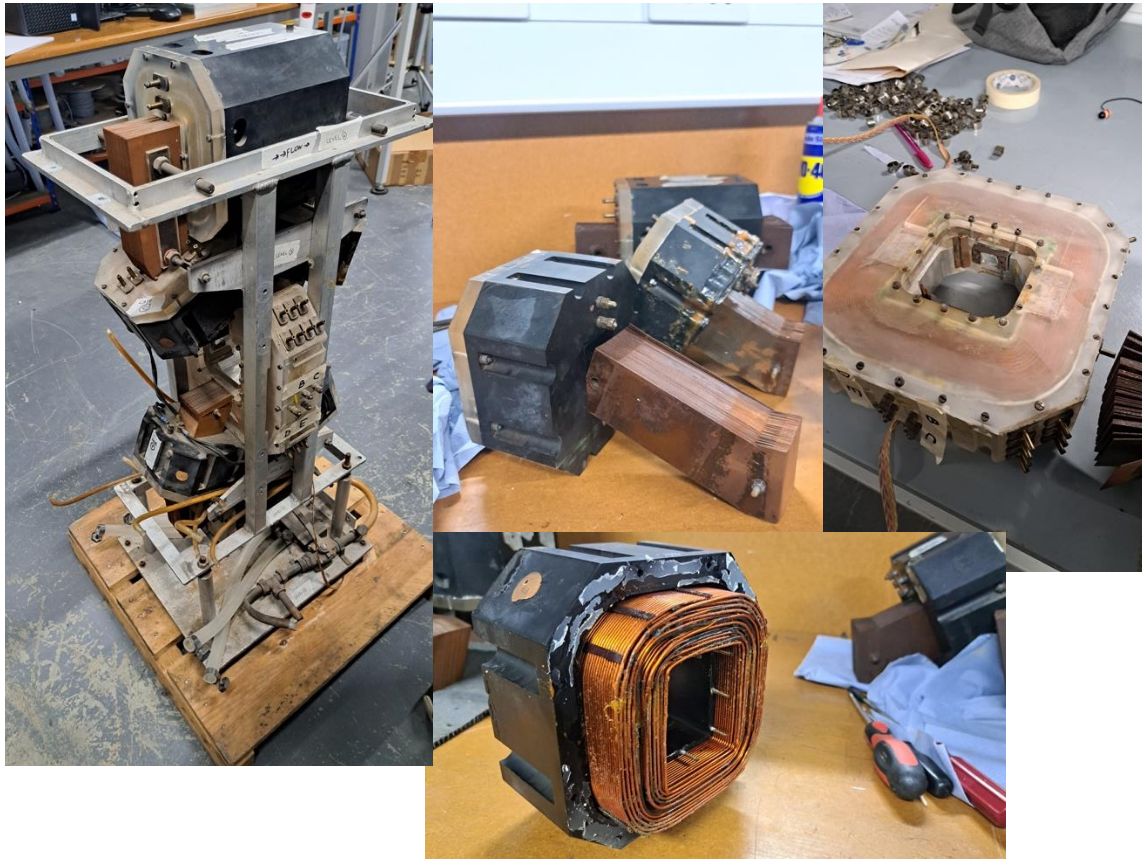


Figure 21: Comparison of CFD simulations between Donaldson's ANSYS study [3] (top), and this project's Eilmer simulations (bottom), presented for the most rarefied case at $\dot{m} = 5\ SLPM$, $T_0 = 385K$, and $p_0 = 872Pa$



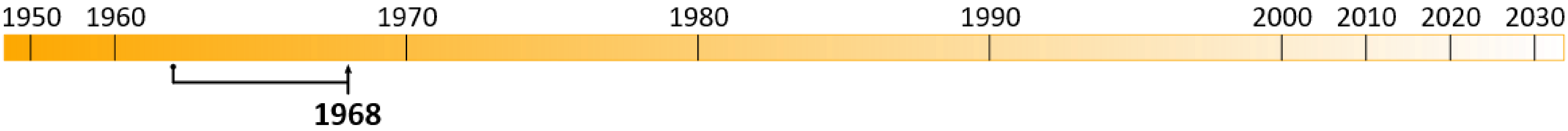
3-DoF MSBS Recommissioning



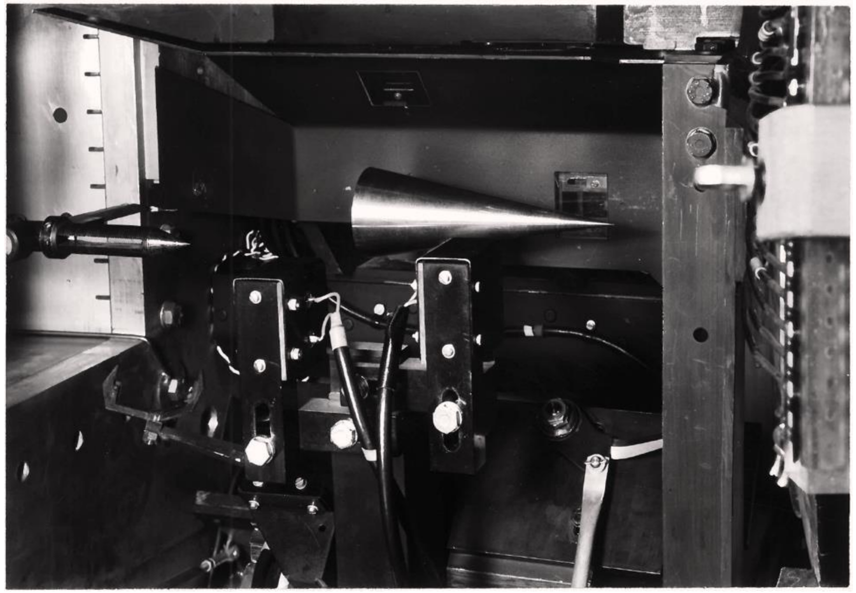


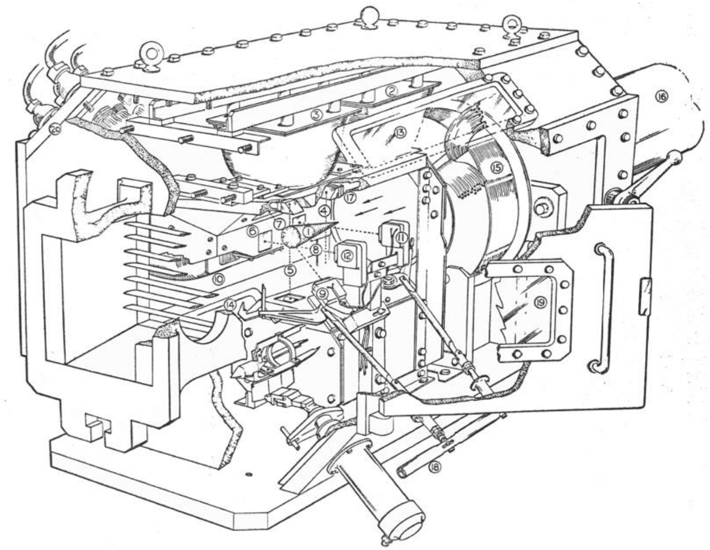
In progress...

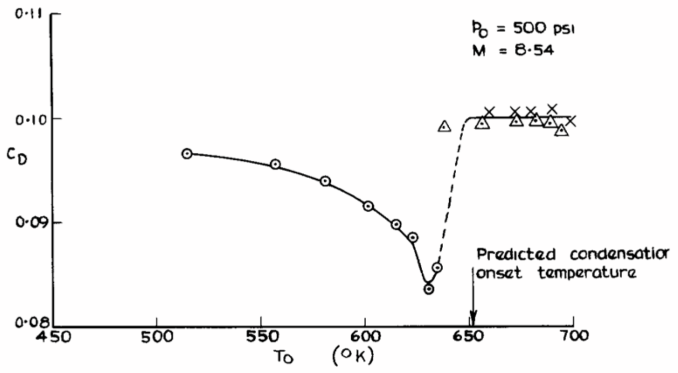
RAE 6-DoF MSBS



- J.F.W. Crane; L.F. Crabtree; J.G. Woodley; A. Wilson; B. Luff
- 7 in. x 7 in. (180 mm) hypersonic wind tunnel
- M ~ 7 – 8.5; T_0 ~ 700 K; P_0 ~ 35 bar; t ~ 1 – 3 mins
- 10° half-angle cone; ~140 mm length





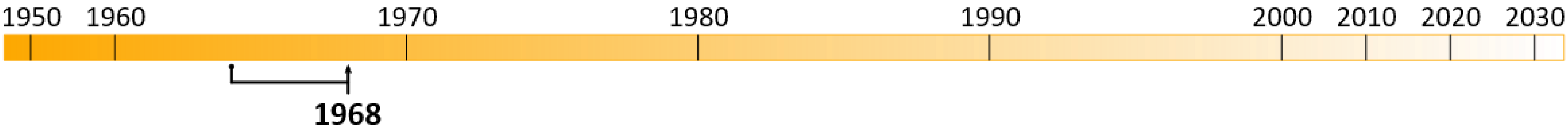


(above) Suspended cone

(left) Schematic drawing of test section

(far left) Influence of condensation on drag

RAE 6-DoF MSBS



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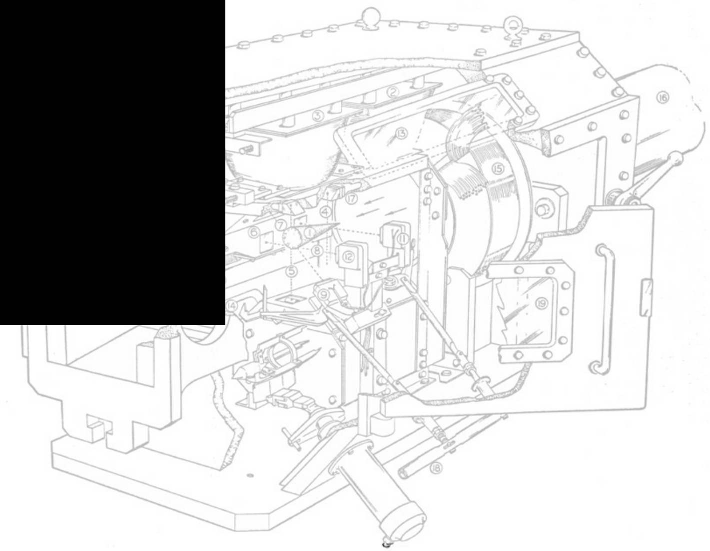

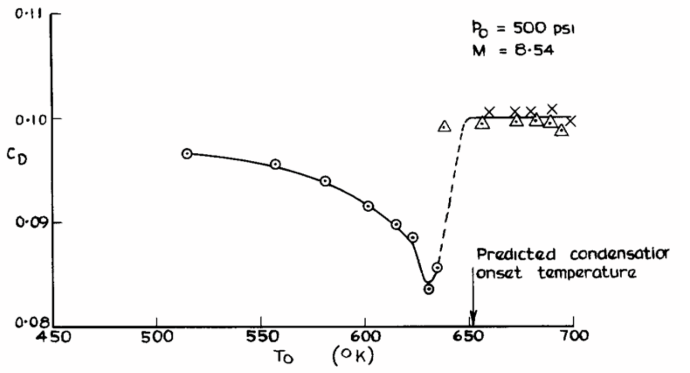

(above) Suspended cone

(left) Schematic drawing of test section

(far left) Influence of condensation on drag

Oxford Hypersonic Facilities Overview

Associate Professor Luke Doherty

Professor Matthew McGilvray (Head of Group)

Associate Professor Tobias Hermann

Mr Chris Hambidge (Facilities Manager)

Oxford Thermofluids Institute

Department of Engineering Sciences, University of Oxford

Contents

UNIVERSITY OF OXFORD

Oxford Thermofluids Institute

- Located 1 mile west of centre of Oxford
- >150 people, incl. 14 academics
- Setup in 1960's by Schultz and Holder
  - Transient wind tunnels and experimentation, particularly to measure heat transfer
- Rolls-Royce University Technology Centre
  - Turbine aerothermal, particulate deposition, seals and electronic cooling
- Large infrastructure allows for large, high TRL facilities
  - Span from TRL 2-7

Oxford Hypersonics Group

- Oxford has over 50 years of hypersonics research heritage
  - Beagle II vehicle experiments (flew in 2003)

- Hypersonics group is led by Prof Matthew McGilvray, created in 2013.
  - Experimental high speed aerothermodynamics
  - US DoD (particularly USAF) support was critical in rapid development
  - MoD UK Centre of Excellence in Hypersonics

- Largest high speed flow research group in the UK (30 members)
  - 2 departmental lecturers (incl. UKRI Future Leaders Fellow), 5 academic collaborators, 6 postdocs and 17 DPhils (incl. 1 USAF Lt, 2 Rhodes scholars, 1 Amelia Earhart Fellow) and 3 full-time tunnel staff.

- Range of fundamental to applied research programmes
  - Diverse range of sponsors (Research Council, Civil and Defence Govn't and Industrial)
  - Lead four multi-million £ grants

Beagle II model being testing in the Oxford Gun Tunnel (decommissioned) at Mach 7.

Oxford Hypersonics Group, March 2022

Oxford Hypersonic Wind Tunnels

NWTF

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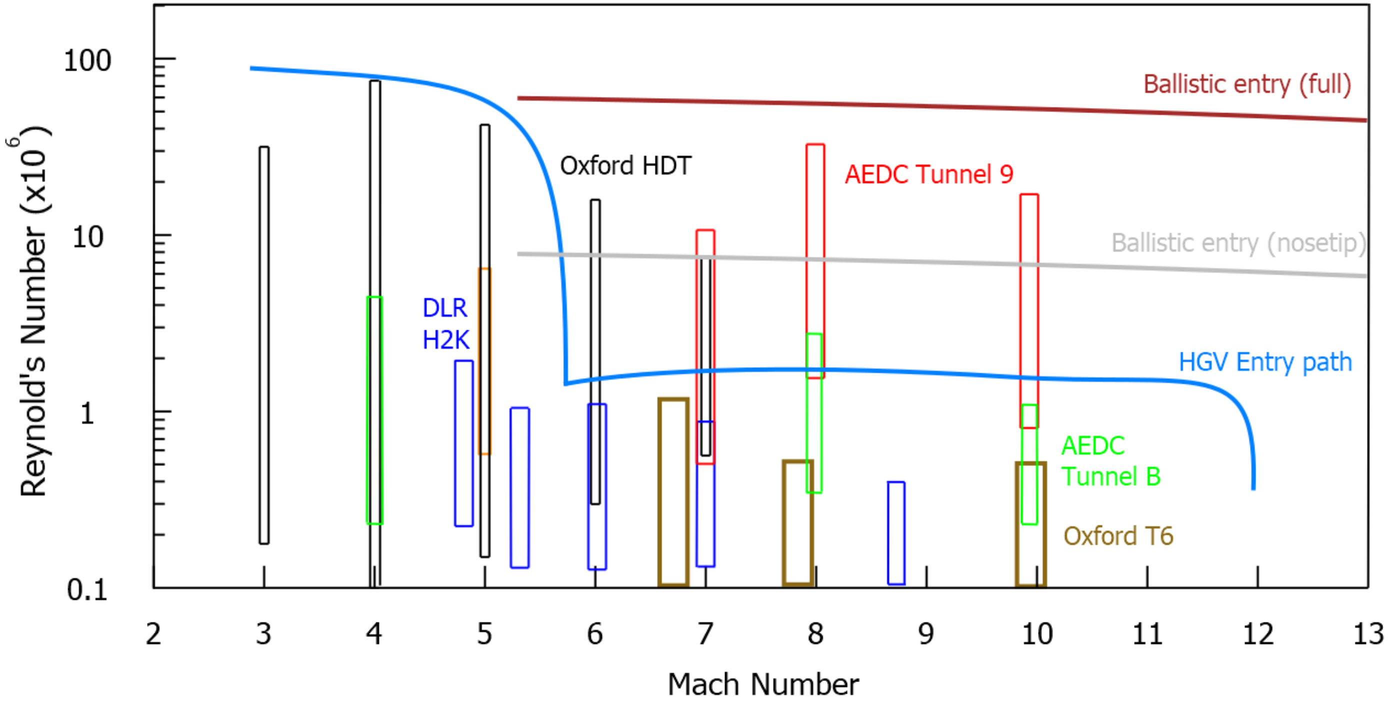
Five (four?) high speed tunnels:

1. Oxford High Density Tunnel
    - High Reynolds number, medium duration
2. Oxford T6 Stalker Tunnel
    - High total enthalpy tunnel
3. Oxford Low Density Tunnel
    - Continuous rarefied tunnel
4. Cold Expansion Tunnel
    - Cold-driver expansion tunnel
5. Osney Plasma Generator
    - Air or N2 continuous plasma torch facility

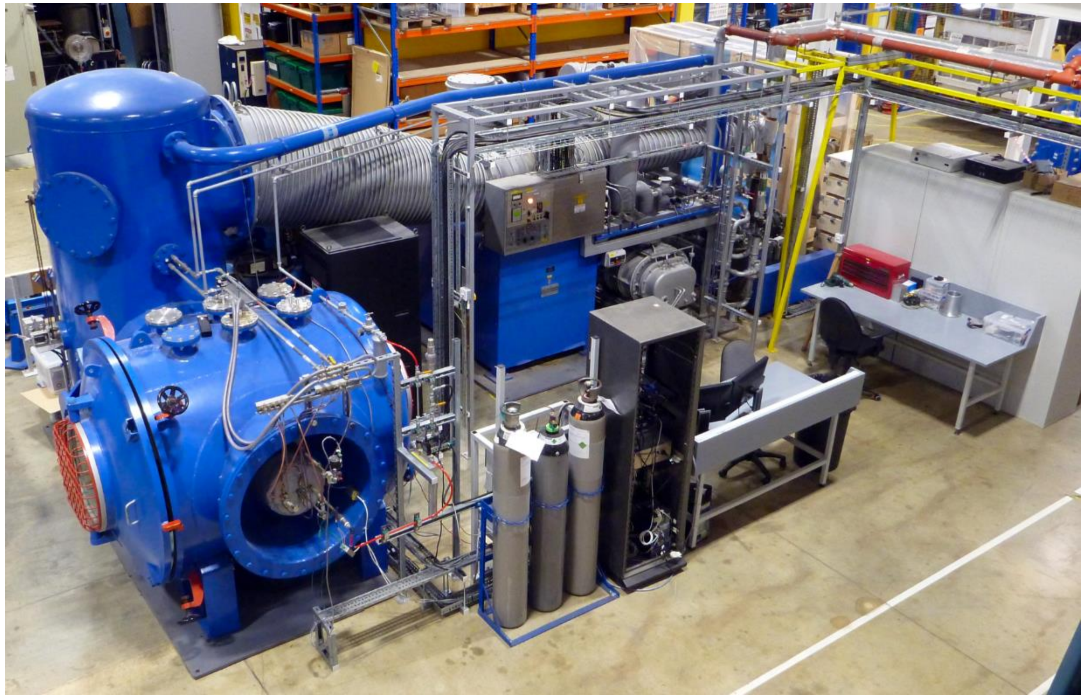


Oxford Hypersonic Wind Tunnels Capability Map

Open access for external research **at least** 25% of year.

Oxford Hypersonic Wind Tunnels

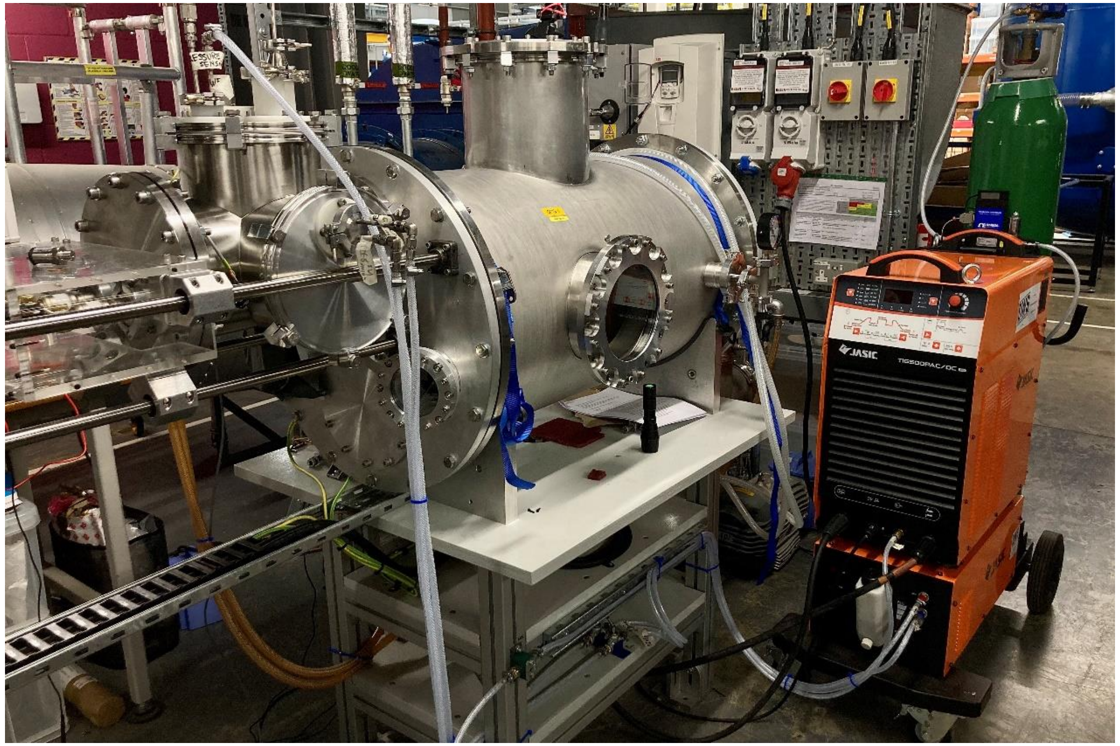
Oxford Hypersonic Laboratory: T6 is on the left and the High Density Tunnel on the right .



Low Density Tunnel

Oxford Hypersonic Wind Tunnels

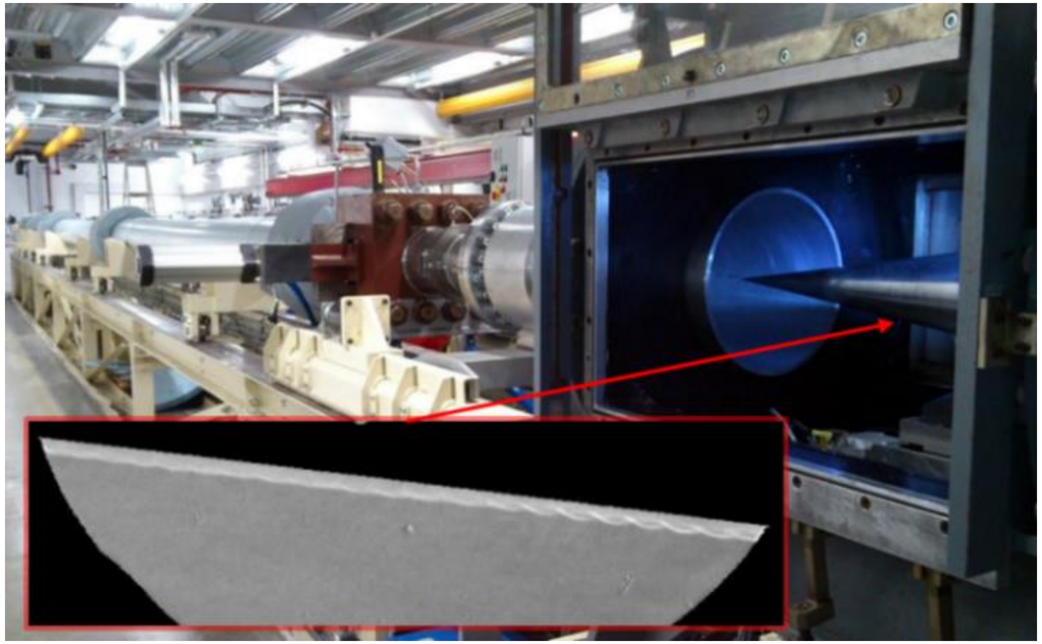
Cold Driven Expansion Tube (now completed).



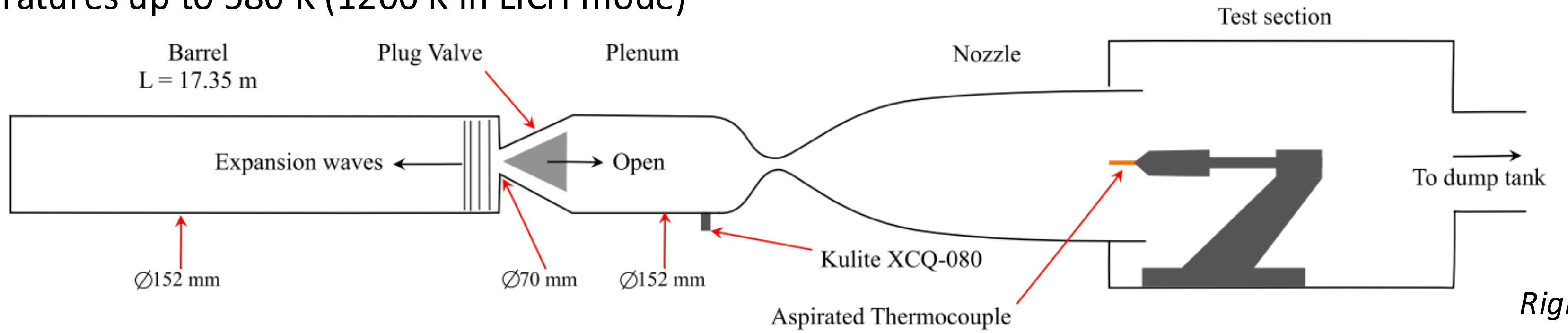
Osney Plasma Generator

High Density Tunnel (HDT)

- Oxford High Density Tunnel was commissioned in 2016
  - Currently undertaken 2700 tests
  - Test times long enough for aerodynamic force measurements (40 – 1000 ms)
  - Large total pressure capability and total temperature capability for a single aerodynamic tunnel
- Plug valve operation → quick turn-around between tests (5 – 10 shots per day)
- Large range of conditions possible
  - Mach 4, 5, 6 and 7 contoured nozzles (all 350 mm exit diameter)
  - Total pressure up to 250 bar (100 bar in LICH mode)
  - Total temperatures up to 580 K (1200 K in LICH mode)



Above: Boundary Layer Instability Experiments on a 7 deg cone



Right: Facility Schematic

High Density Tunnel (HDT)

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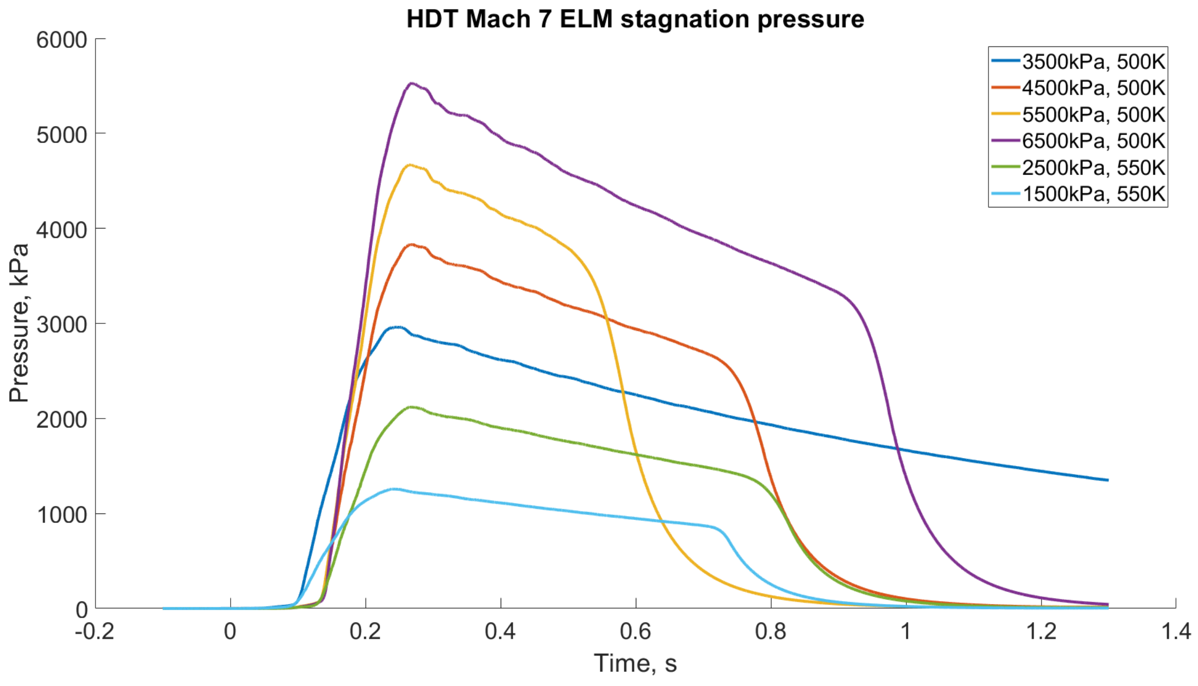
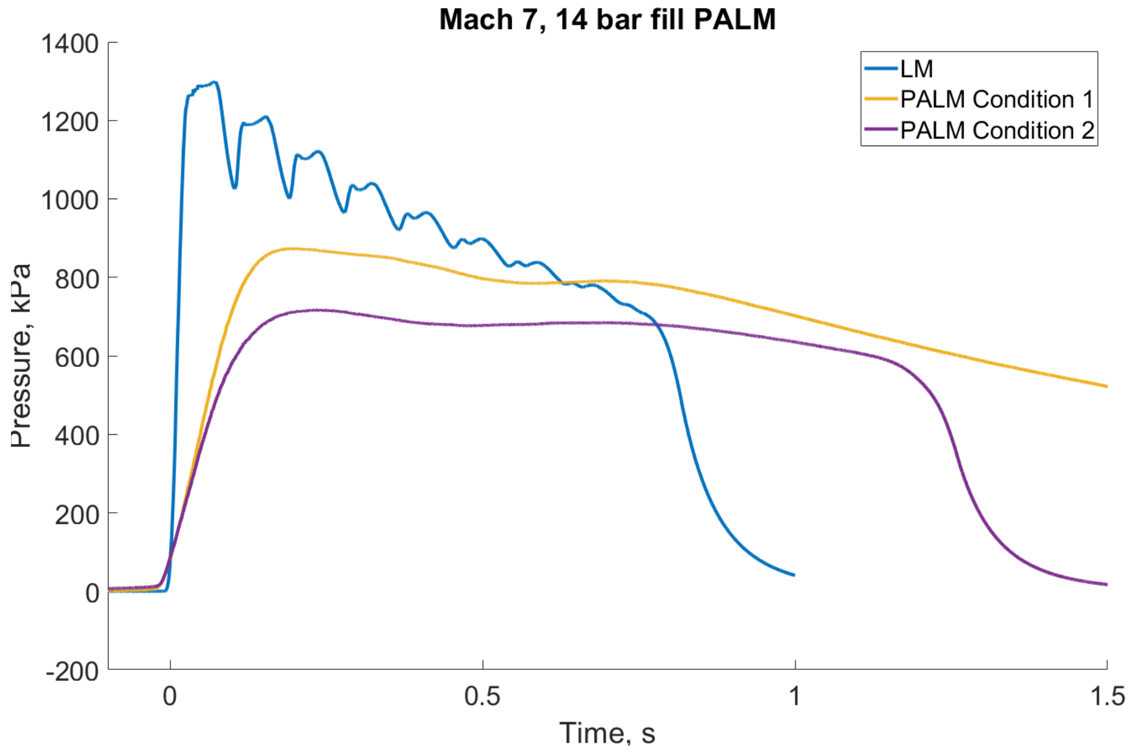
- Developing new long-duration modes → Extended Ludwieg Mode (ELM and Plenum Augmented Ludwieg Mode (PALM)
  - Offers increased test time that is steadier than any other facility in the UK
  - Trade-off Reynolds number capability for test time

Below charts → data from June 2024 (unpublished/confidential)





HDT Experimental Capabilities

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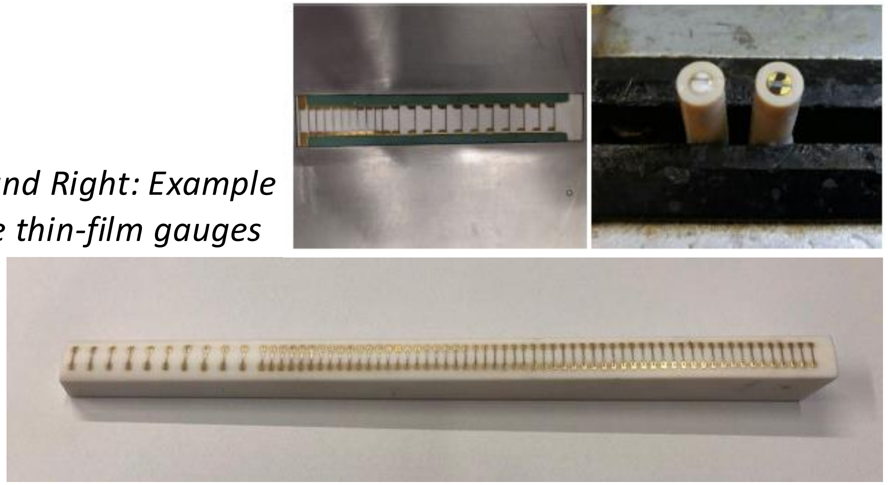
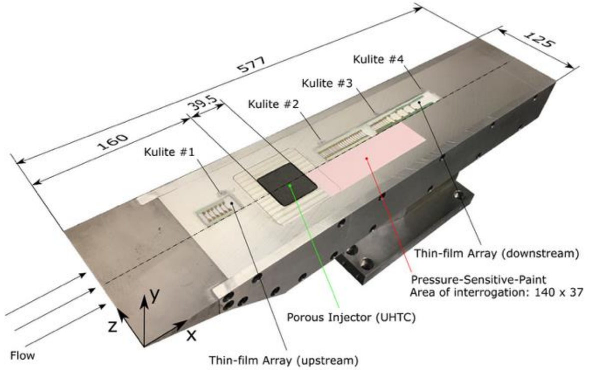
- High-speed Schlieren videography. Cameras available:
  - Phantom TMX 7510: 76 kfps @ 1280 x 800 → 1.75 Mfps @ 1280 x 32
  - Specialised Imaging Kirana 5M: 5 Mfps @ 924 x 768
  - Photron FASTCAM Mini AX-200: 6.4 kfps @ 1024 x 1024 → 900 kfps @ 128 x 16
  - Photron FASTCAM Mini UX-100: 4 kfps @ 1280 x 1024 → 800 kfps @ 640 x 8
- High-speed (Mid-Wave) Infrared Thermography. Camera:
  - Telops FAST M3K: 3.1 kfps @ 320 x 256 → 1 Mfps @ 64 x 4
- In-house bespoke manufacturing of thin-film heat transfer gauges
- Focussed Laser Differential Interferometry (FLDI)
- Laser Induced Grating Spectroscopy (LIGS) for Optical Thermometry
- Free-Flight force measurement capability
- Pressure Sensitive Paint (single and two colour)

- Fast response pressure sensors (Kulite and PCB Piezotronics)
- Bespoke inlet mass capture measurement and back-pressure device (MassCap)

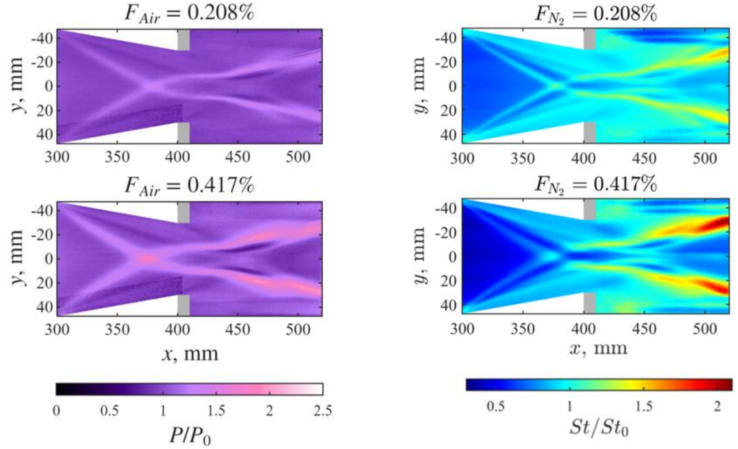
Below and Right: Example bespoke thin-film gauges



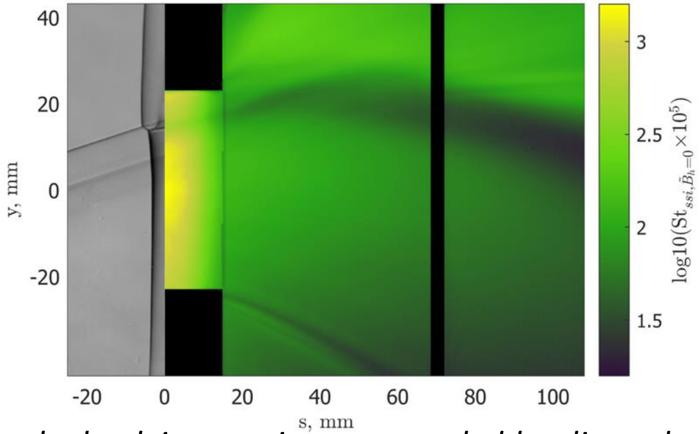
HDT Research Examples



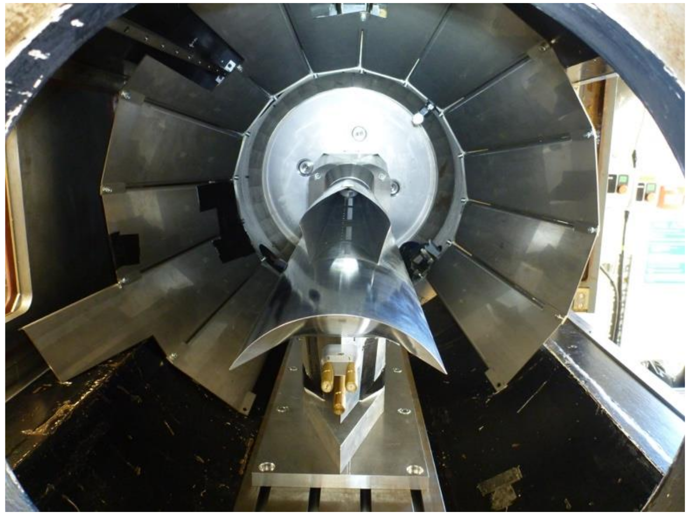
Transpiration cooling on an instrumented flat plate



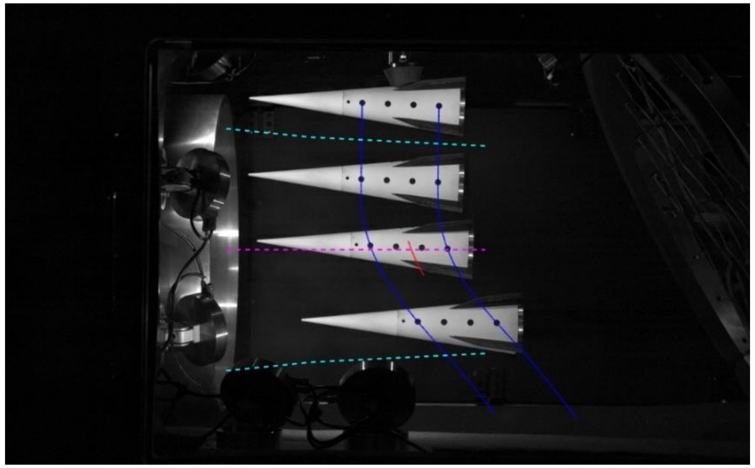
Swept shock-shock boundary layer interaction with cooling



Shock-shock interaction on a cooled leading edge



Scramjet inlet mass capture & self-starting capability



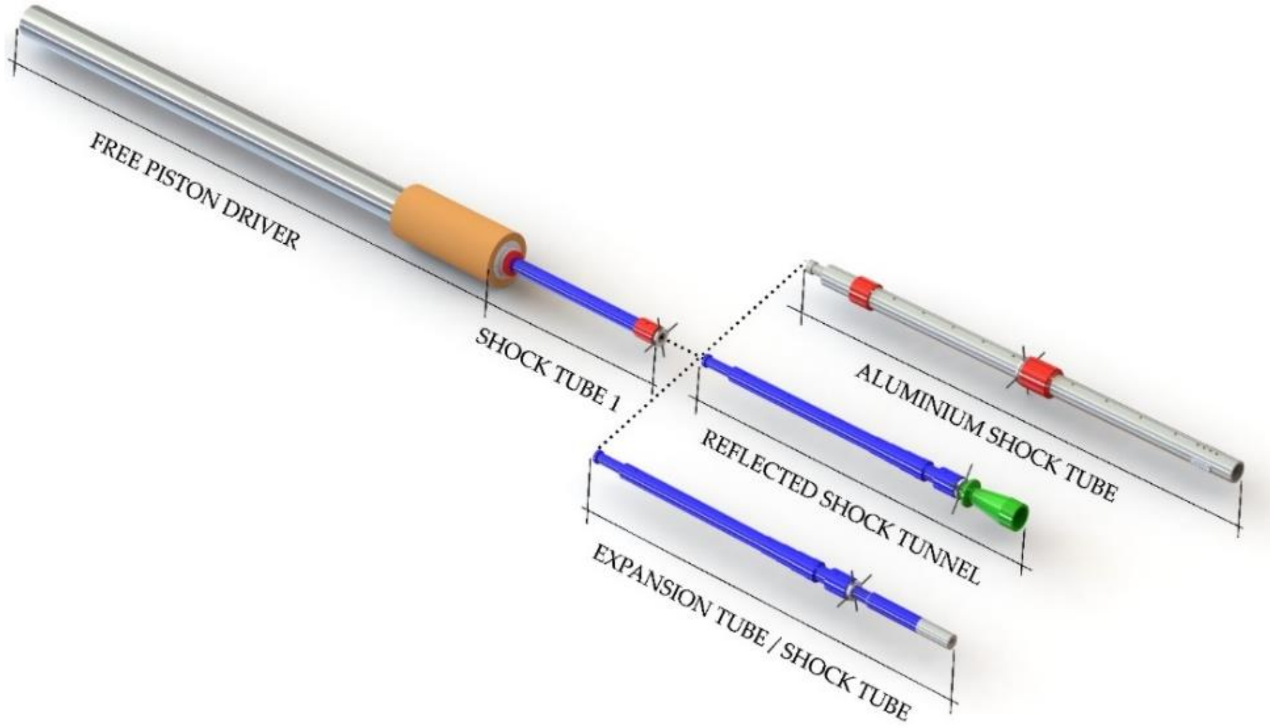
Free-flight aerodynamic force measurement of a winged cone (Mach 7)

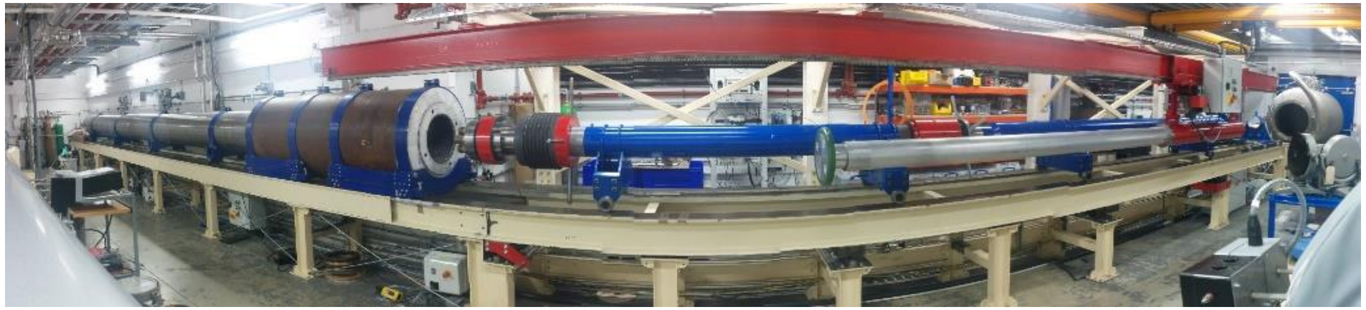
Roll-controlled cone at Mach 7

T6 Stalker Tunnel

- Operates as a reflected shock tunnel, expansion tunnel or shock tube
  - Allows for one model to be tested across a large range of conditions and one set of instrumentation.
  - Can move from conditions in reflected shock tunnel mode to expansion tunnel mode to overcomes chemically dirty/total pressure issues at expense of test time
  - Shock tube mode overcomes difficulties of scaling radiation with sub-scaling of vehicle to fit in wind tunnels.
  - Mach 7, 8, 10 nozzles; exit dia. 96 – 280 mm
  - 1,500 K < T_0 < 20,000 K;   10 MPa < P_0 < 1000 MPa
  - Flow durations 0.01 ms → 10 ms

- Commissioned in 2018, has undertaken over 400 tests across all modes.
  - Only shock tube and expansion tube in Europe



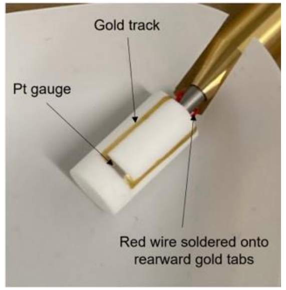


T6 Experimental Capabilities

- Visible, near-infrared spectroscopy:
  - Princeton Instruments Isoplane 320 with an Andor sCMOS camera
- Rated for use with H2 as test gas (gas-giant entry, shock layer radiation experiments)
- High-speed Schlieren videography. Cameras available:
  - Phantom TMX 7510: 76 kfps @ 1280 x 800 → 1.75 Mfps @ 1280 x 32
  - Specialised Imaging Kirana 5M: 5 Mfps @ 924 x 768
- High-speed (Mid-Wave) Infrared Thermography. Camera:
  - Telops FAST M3K: 3.1 kfps @ 320 x 256 → 1 Mfps @ 64 x 4
- In-house bespoke manufacturing of thin-film heat transfer gauges
- Fast response pressure sensors (Kulite and PCB Piezotronics)
- Focussed Laser Differential Interferometry (FLDI)

In-development:

- Free-Flight force measurement capability
- Combustible gas injection system
- Vacuum ultraviolet spectroscopy:
  - McPherson 207V with an Andor sCMOS camera
- Plasma pre-heating of models



Left: Example bespoke thin-film gauges on Macor for satellite-demise heat transfer

Right: Diamond heat transfer gauge

T6 Research Examples

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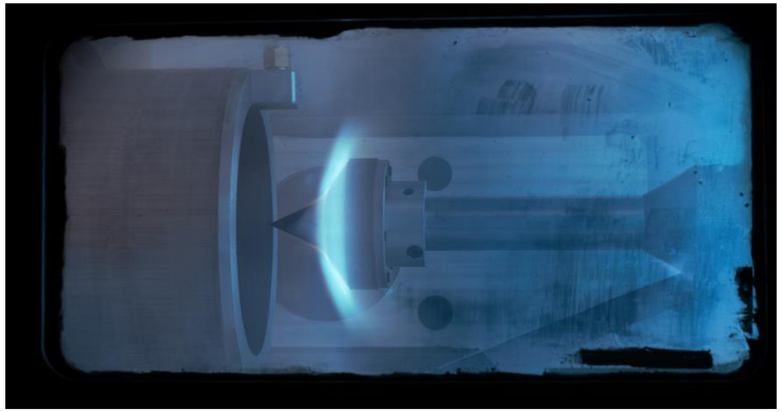
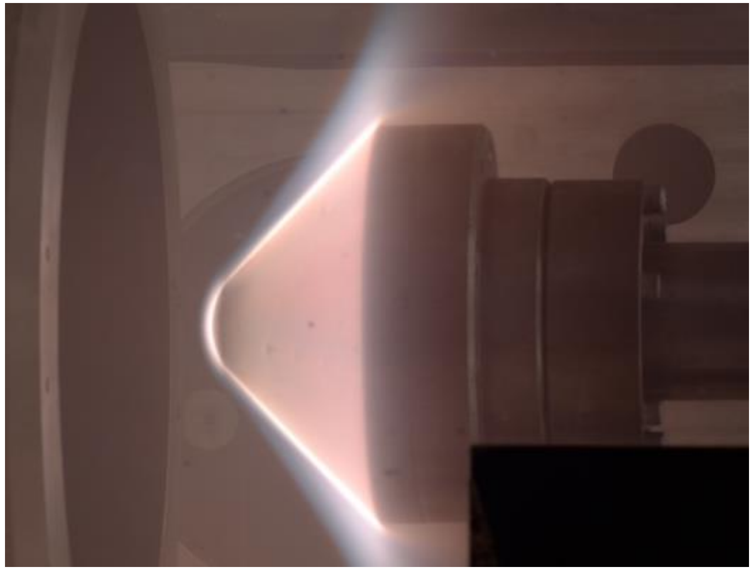
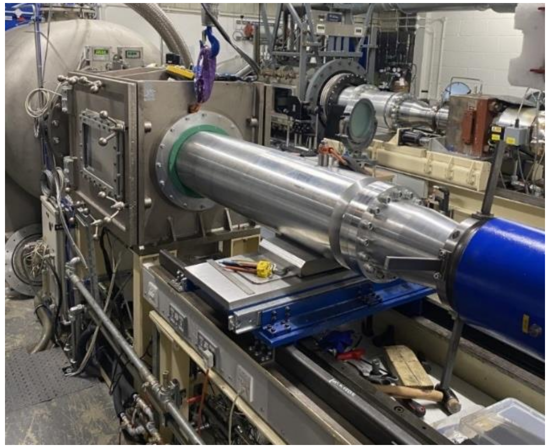


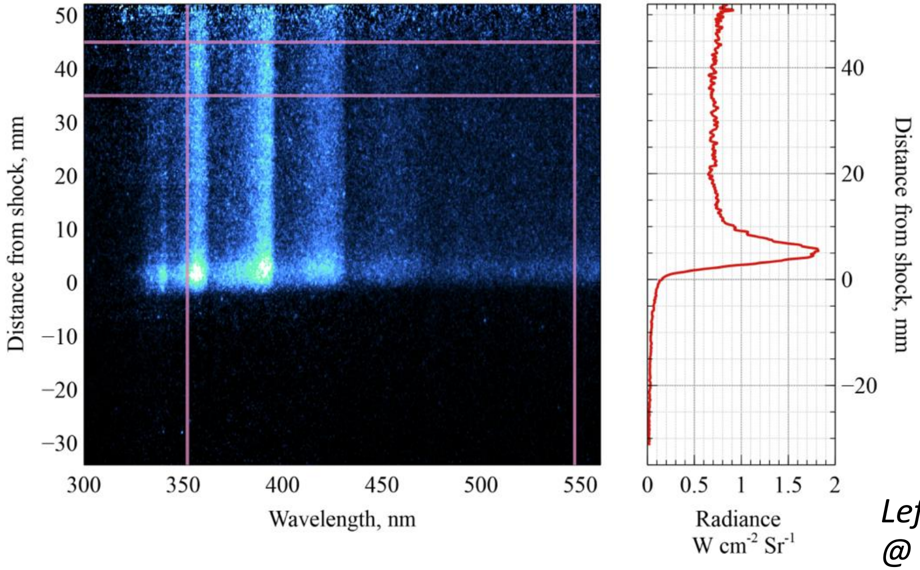
Flow luminosity: double-cone model tested in Expansion Tunnel mode at 6.3 km/s (19.8 MJ/kg) for earth entry.



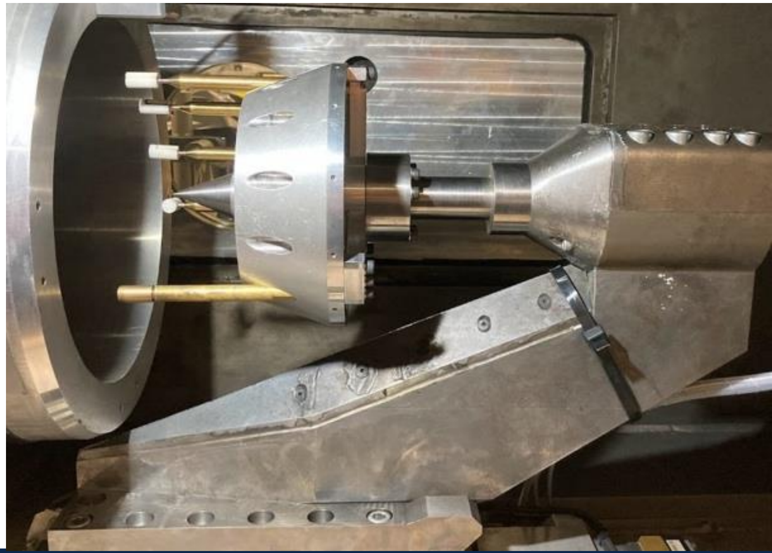
Flow luminosity: sub-scale Galileo model tested at 18.0 km/s for Neptune entry (84.5% H_2 / 15% He / 0.5% CH_4).



Above: Contoured Mach 10 expansion nozzle. Exit dia. of 236 mm



Left: Example air spectra at 10 km/s @ 58 km altitude flight condition.



Right: Satellite demise heat transfer testing at 6.5 km/s for LEO return.

Low Density Tunnel (LDT)

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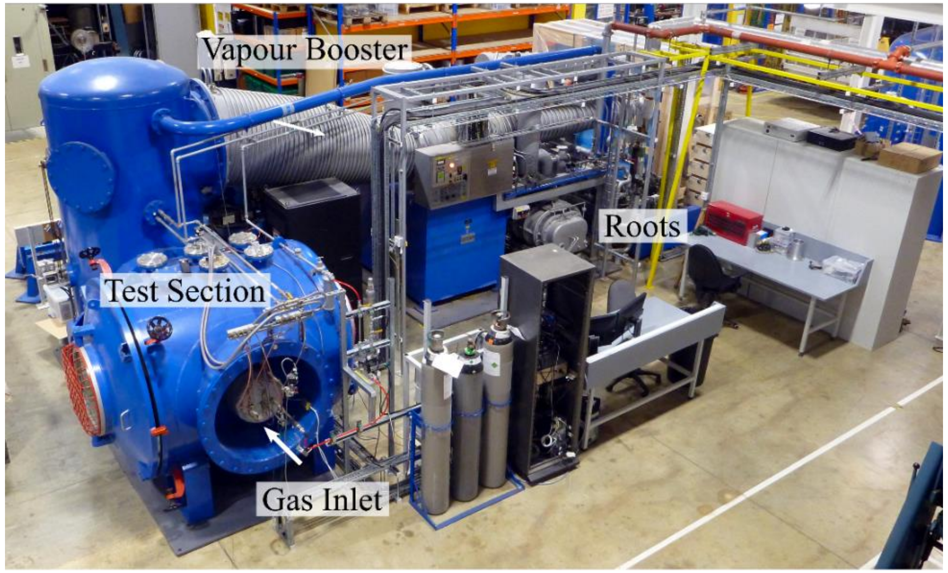
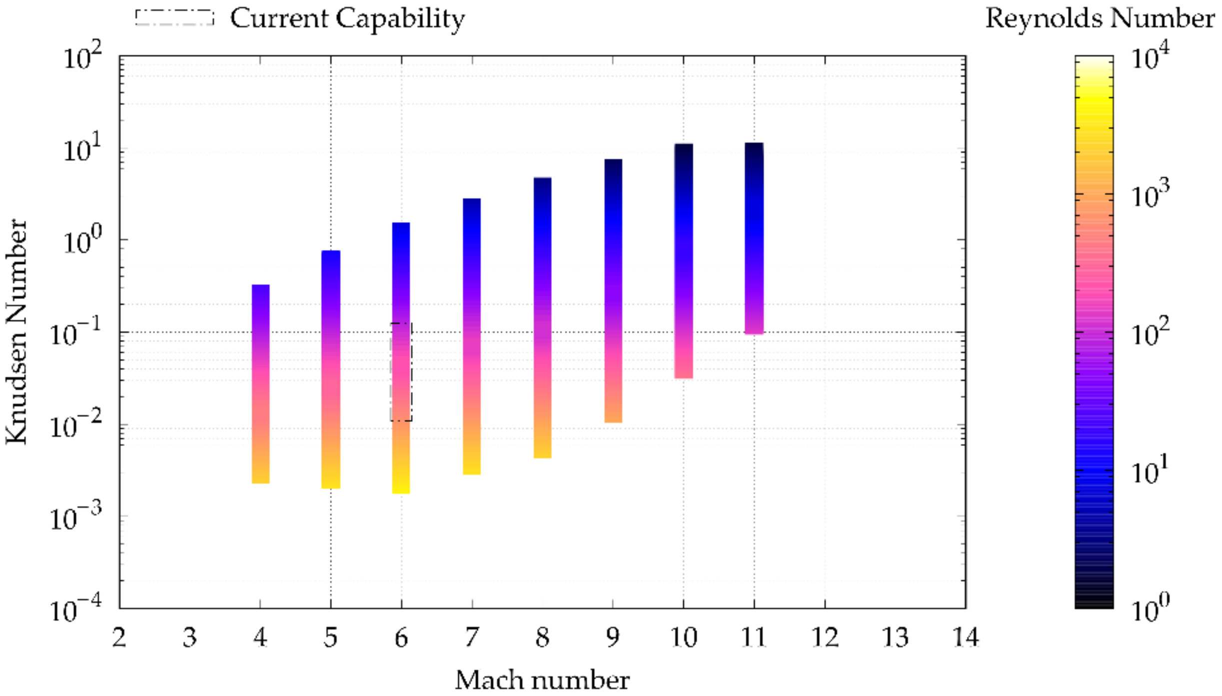
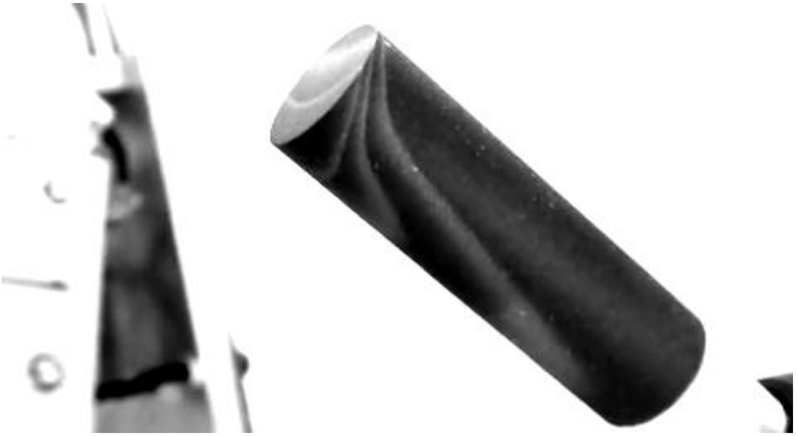
- Continuous facility; 3-stage vacuum pumping system; w_{max} = 0.6 g/s.
- Test gas: dry air; any inert bottled gas (N2, CO2, etc.)
- Test Section 1.5 m diameter
- Total temperatures up to 523 K
- Mach 6 contoured nozzle; exit diameter 108 mm



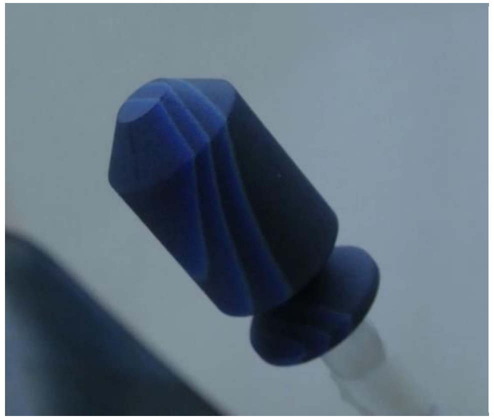


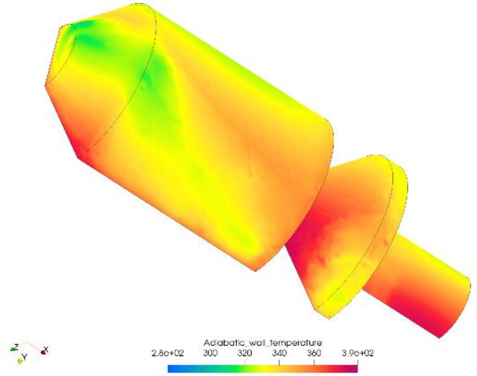
- Thermochromic Liquid Crystal Thermography for Heat Transfer Measurements
- Magnetic suspension balance (MSBS; undergoing refurbishment).
- Linear (3-axis) and rotary (2-axis) traverse systems.



(Left) Raw image of cylinder model coated with 3-temperature liquid crystal slurry at Ma = 45° test condition.

- Prior research using the LDT includes:
  - (Recent) Edge heating effects on space debris surrogate geometries.
  - Wind sensors (anemometer) for Mars landers.
  - Aerodynamics of cones and heat shields using the MSBS.
  - Heat transfer studies on cones and spheres.

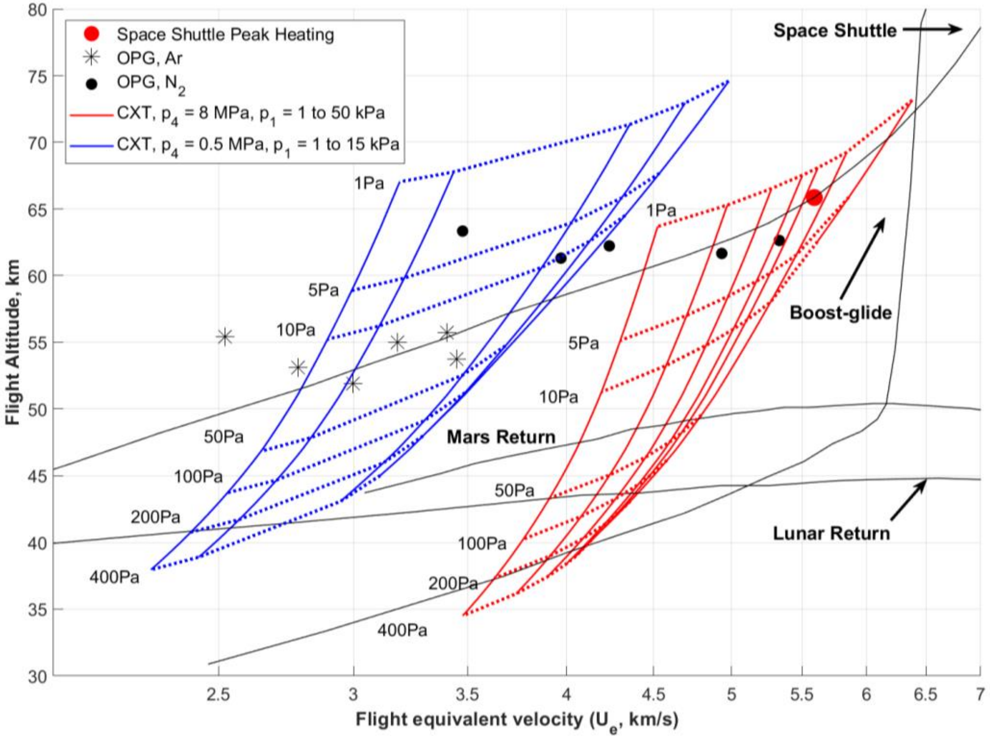


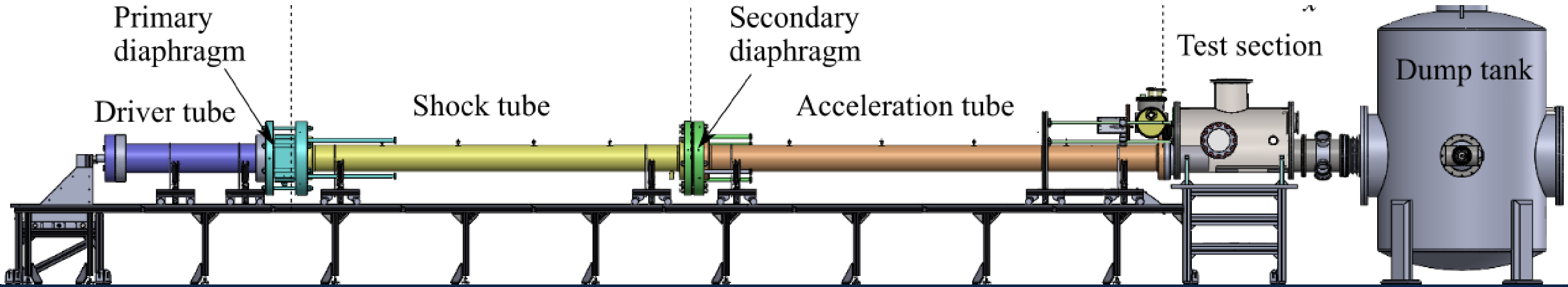


(above left) Raw camera image of upper-stage satellite analogue coated with a 3-temperature liquid crystal slurry at Ma = 5.8, Kn ≈ 0.02. (above right) Calculated adiabatic wall temperature.

Cold-Driven Expansion Tube (CXT)

- Aiming at 2~7 km/s for integrated model preheating (via. OPG) and hypervelocity testing
- Condition enthalpy 11 MJ/kg, equivalent flight velocity ~4.7 km/s
- Now commissioned

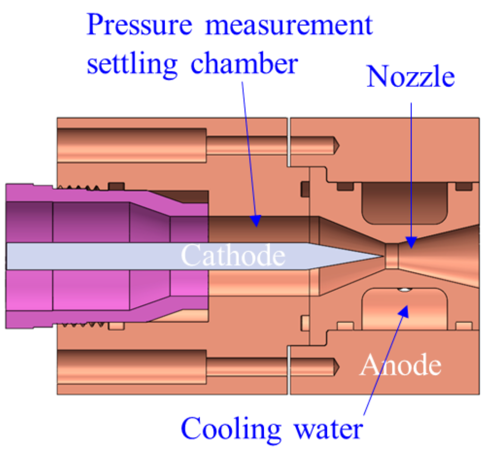




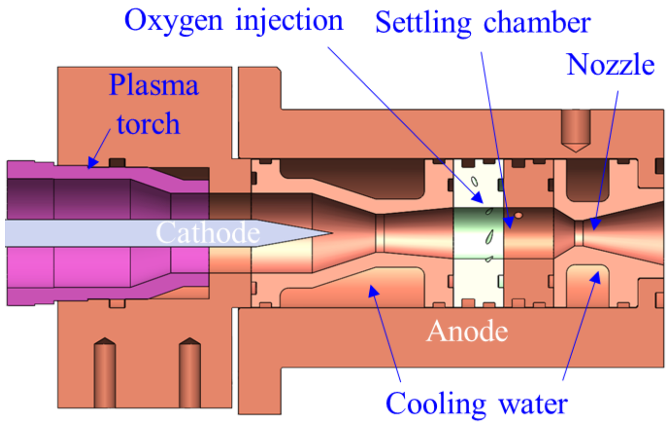
Osney Plasma Generator (OPG)

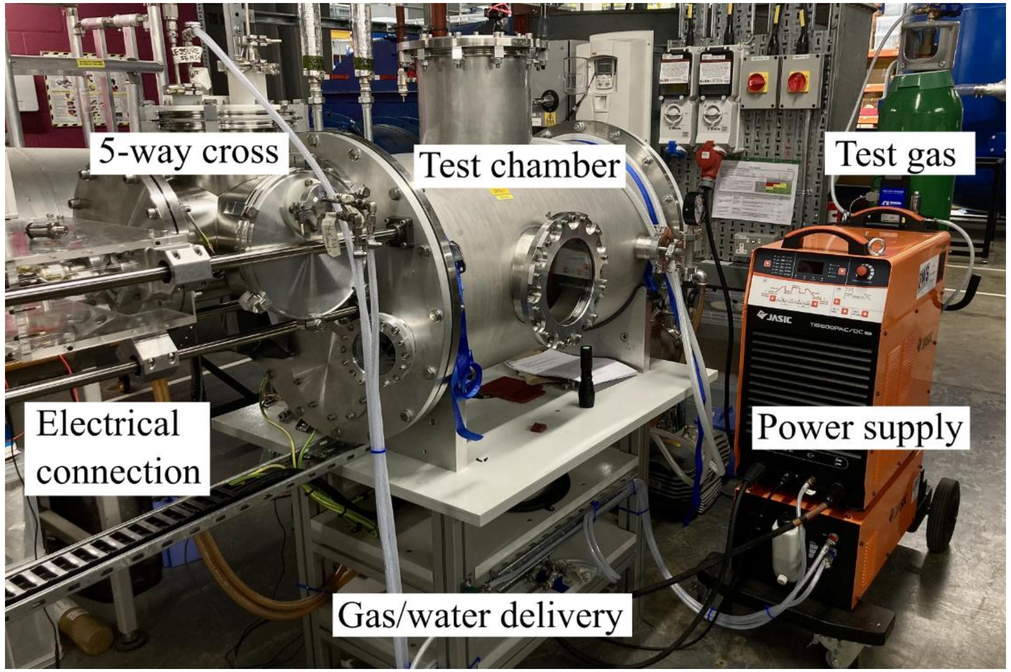
- Thermal Arc-jet facility
- Power supply: Tungsten Inert Gas (TIG) welding power supply, 21.5 kW, 500 A DC
- Test gases: Nitrogen, Argon, Air at mass flow rates up to 1 g/s
- First run: 14th April 2023

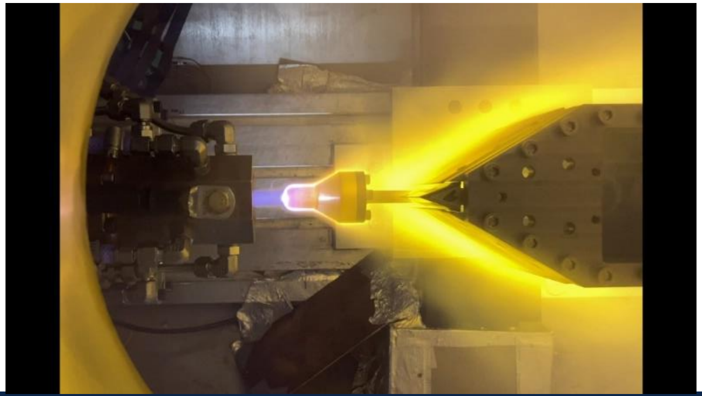
OPG1



OPG2

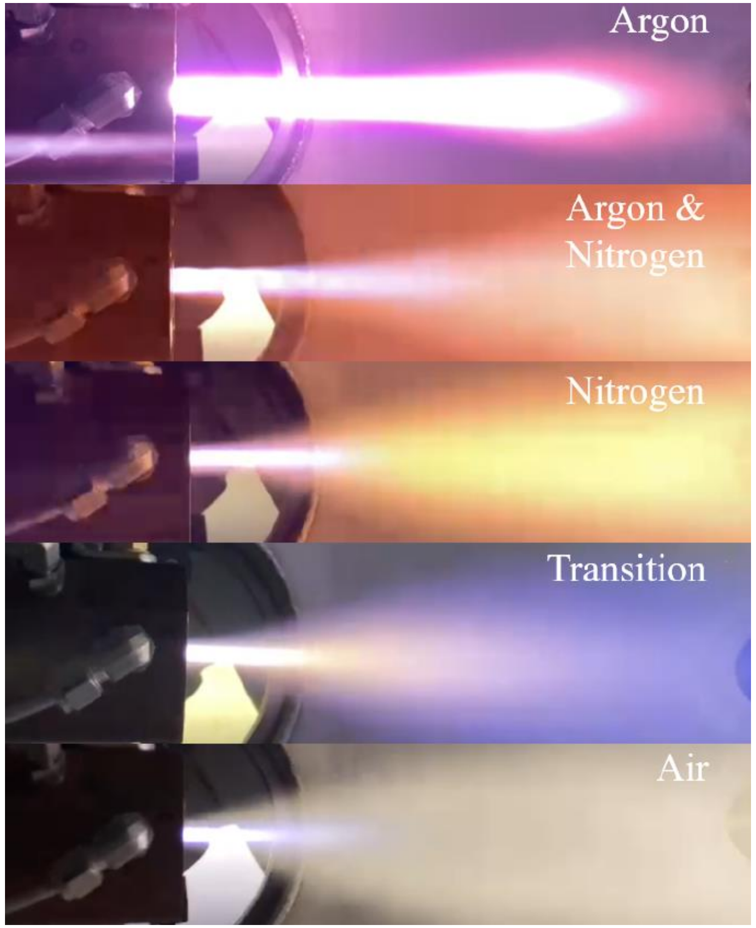






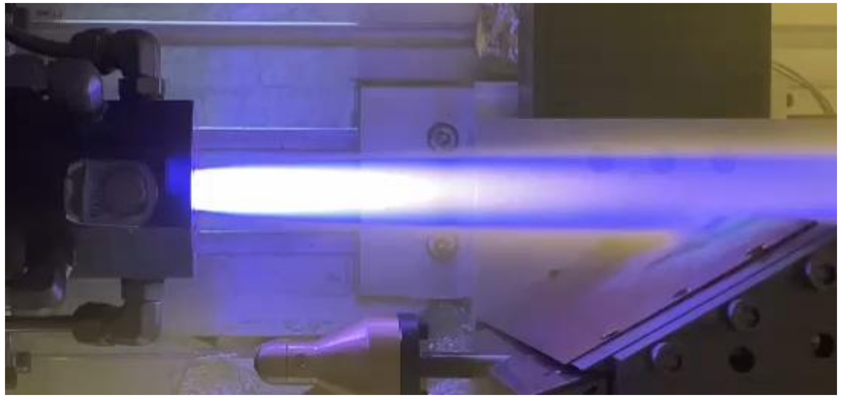
OPG Recent Experiments

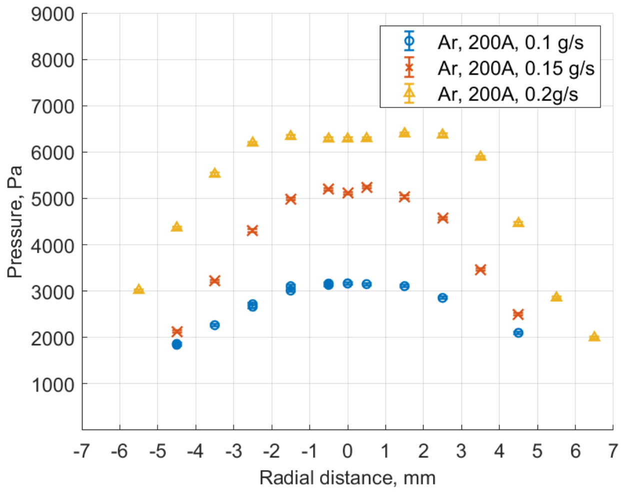
- Primarily facility characterisation and run-up



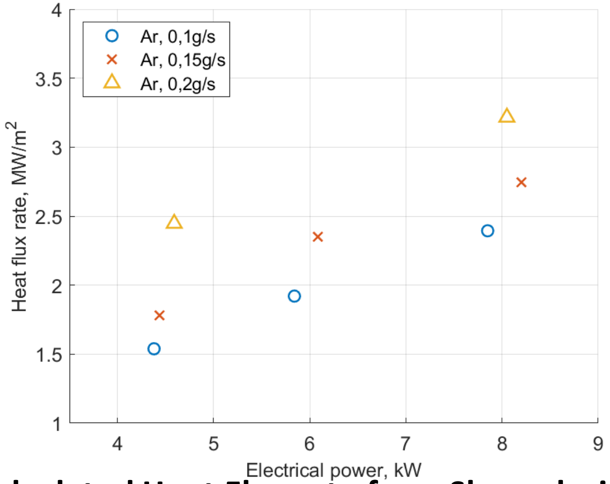
Slug calorimeter

Nitrogen, 400A, 0.07g/s





Pitot (stagnation) pressure profile



Calculated Heat Flux rate from Slug calorimeter

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- Oxford Thermofluids Institute: https://oti.eng.ox.ac.uk/
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- Mr Chris Hambidge (Hypersonic Facilities Manager)

  E: chris.Hambidge@eng.ox.ac.uk W: https://oti.eng.ox.ac.uk/people/chris-hambidge/

HDT Selected Research Papers

HDT Facility Commissioning

1. Wylie, S., Doherty, L.J., McGilvray, M. (2018), “Commissioning of the Oxford High Density Tunnel for Boundary Layer Stability Measurements at Mach 7”, *48th AIAA Fluid Dynamics Conference*, DOI: https://doi.org/10.2514/6.2018-3074
2. McGilvray, M., Doherty, L., Neely, A.J., Ireland, PTI, and Pearce, R. (2015), “The Oxford High Density Tunnel,” *20th AIAA International Space Planes and Hypersonic Systems and Technologies Conference*, DOI: https://doi.org/10.2514/6.2015-3548.

IR Thermography

3. Naved, I., Hermann, T., McGilvray, M., Ewenz Rocher, M., Hambidge, C., Doherty, L.J., Le Page, L., Grossman, M., Vandeperre, L. (2022), “Heat Transfer Measurements of a Transpiration Cooled Stagnation Point in Transient Hypersonic Flow”, *Journal of Thermophysics and Heat Transfer*, DOI: https://doi.org/10.2514/1.T6610

Free-flight Force Measurements

4. Hyslop, A., Doherty, L.J., McGilvray, M. (2022), “Free-Flight Aerodynamic Testing of a 7 Degree Half-Angle Cone”, *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2022-1324

Pressure Sensitive Paint

5. Ifti, H.S., Hermann, T., Ewenz Rocher, M., Doherty, L.J., Hambidge, C., McGilvray, M., Vandeperre, L. (2022), “Laminar Transpiration Cooling Experiments in Hypersonic Flow”, *Experiments in Fluids*, vol. 63:102, DOI: https://doi.org/10.1007/s00348-022-03446-1
6. Wheeler, C., Hyslop, A., Vieira, J., Le Page, L., Quinn, M.K., Chowdhury, N.H.K., Doherty, L.J. (2022), “Surface Pressure Measurements on a Free-Flying Cone at Mach 7 using Pressure Sensitive Paint”, *2nd HiSST Conference*, URL: https://ora.ox.ac.uk/objects/uuid:87cd990a-b664-464f-b9ea-447e2c127167
7. Ewenz Rocher, M., Hermann, T., McGilvray; Ifti, H. S., Vieira, J., Hambidge, C., Quinn, M., Grossman, M., Vandeperre, L. (2022), "Pressure Sensitive Paint Diagnostic to Measure Species Concentration on Transpiration Cooled Walls", *Experiments in Fluids,* vol. 63:2, DOI: https://doi.org/10.1007/s00348-021-03355-9

High-Speed Inlet Mass Capture

8. Doherty, L.J., Hambidge, C., Ivison, W., Hyslop, A., McGilvray, M. (2019), “Development of a Novel Mass Flow Measurement Device for High Speed Inlet Testing”, *Int’l Conference on Flight Vehicles Aerothermodynamics and Re-entry Missions & Engineering (FAR)*, url: https://ora.ox.ac.uk/objects/uuid:f5c7608a-fc89-4267-aa77-2f8a275fc55f

Heat Transfer Measurements

9. Ivison, W., Hambidge, C., McGilvray, C., Merrifield, J., Steelant, J. (2022), “Experimental Investigation of the Effect of Steps and Gaps on Hypersonic Vehicles”, *2nd HiSST Conference*, url: https://ora.ox.ac.uk/objects/uuid:eea4bd8d-95c8-4c3d-b244-cec147d216b1

High-Speed Schlieren

10. Kerth, P., Wylie, S., Ravichandran, R. and McGilvray, M. (2022), “Gas Injection into Second Mode Instability on a 7 Cone at Mach 7,“ *AIAA Aviation Forum*, DOI: https://doi.org/10.2514/6.2022-3856

T6 Selected Research Papers

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T6 Facility Commissioning

1. Collen, P., Doherty, L.J., Subiah, S.D., Sopek, T., Jahn, I., Gildfind, D., Penty Geraets, R., Gollan, R., Hambidge, C., Morgan, R. McGilvray, M. (2021), "Development and Commissioning of the T6 Stalker Tunnel", *Experiments in Fluids*, vol. 62, article 225, DOI: https://doi.org/10.1007/s00348-021-03298-1
2. Subiah, S.D., Collen, P.L., Doherty, L.J., Penty Geraets, R., Hyslop, A., McGilvray, M. (2019), "Condition Development and Commissioning of the Oxford T6 Stalker Tunnel in Reflected Shock Tunnel Mode", *Int'l Conference on Flight Vehicles Aerothermodynamics and Re-entry Missions & Engineering (FAR),* URL: https://ora.ox.ac.uk/objects/uuid:706a8d91-bc67-4779-ae96-aa3be3549551
3. McGilvray, M., Collen, P., Doherty, L.J., Steer, J., Leader, J., Glenn, A. and Hambidge, C. (2022), "The Oxford T6 Stalker Tunnel: Performance, Upgrades & New Modes of Operation," *Int'l Conference on Flight Vehicles Aerothermodynamics and Re-entry Missions & Engineering (FAR)*, URL: https://ora.ox.ac.uk/objects/uuid:a792a023-e907-4313-bbb7-0bb3754f2f02
4. Chang, E. W. K., Hermann, T., McGilvray, M., James, C.M. (2022) "Mars Sample Return Flow Condition Design and Pitot Rake Testing in T6 Stalker Tunnel", *9th Int'l Workshop on the Radiation of High Temperature Gases for Space Mission (RHTG-9)*.

Shock Layer Radiation Experiments

5. Collen, P., Satchell, M., Di Mare, L., McGilvray, M. (2022) "The influence of shock speed variation on radiation and thermochemistry experiments in shock tubes," *Journal of Fluid Mechanics* Vol. 948, A51, DOI: https://doi:10.1017/jfm.2022.727
6. Collen, P., Glenn, A., Doherty, L.J., McGilvray, M. (2022), "Measurements of Absolute Radiative Emission from Hypervelocity Air Shocks in the T6 Aluminium Shock Tube," *Journal of Heat Transfer and Thermophysics* (under review)
7. Collen, P.L., Doherty, L.J., McGilvray, M. (2019), "Measurements of Radiating Hypervelocity Air Shock Layers in the T6 Free-Piston Driven Shock Tube", *Int'l Conference on Flight Vehicles Aerothermodynamics and Re-entry Missions & Engineering (FAR)*, URL: https://ora.ox.ac.uk/objects/uuid:86901f15-a619-499c-8099-0f601f627ad6
8. Glenn, A.B., Collen, P.L., McGilvray, M. (2022), "Experimental Non-Equilibrium Radiation Measurements for Low-Earth Orbit Return," *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2022-2154
9. Steer, J., Collen, P.L., Glenn, A., Hambidge, C., Doherty, L.J., McGilvray, M., Loehle, S., Walpot, L., (2022), "Shock Radiation Tests for Ice Giant Entry Probes including CH4 in the T6 Free-Piston Driven Wind Tunnel", *9th Int'l Workshop on the Radiation of High Temperature Gases for Space Mission (RHTG-9),* url: https://ora.ox.ac.uk/objects/uuid:8c443bc7-8e78-4d5d-8f2b-526ed58674dc

Heat Transfer Measurements

10. Steer, J., Collen, P., Glenn, A., Sopek, T., Hambidge, C., Doherty, L.J., Loehle, S., Walpot, L. and McGilvray, M. (2022), "Experimental simulation of a Galileo sub-scale model at Ice Giant entry conditions in the T6 free-piston driven wind tunnel," *9th Int'l Workshop on the Radiation of High Temperature Gases for Space Mission (RHTG-9)*, URL: https://ora.ox.ac.uk/objects/uuid:7665ccca-14aa-4d2a-95c3-57cb6e9d41dc
11. Leader, J., Steer, J., Collen, P., McGilvray, M. (2022), "Measuring Re-Entry Aeroheating of Satellite Components at True Fligth Total Enthalpies", *9th Int'l Workshop on the Radiation of High Temperature Gases for Space Mission (RHTG-9),* URL: https://ora.ox.ac.uk/objects/uuid:dd48a65c-da9a-4074-865b-d5cf01d8b560

LDT Selected Research Papers

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LDT Facility Commissioning

1. Donaldson, N.L., Doherty, L.J., Ivison, W., Wilson, C.F., McGilvray, M., Ireland, P.T. (2019), “Refurbishment and Characterisation of the Oxford Low Density Hypersonic Wind Tunnel”, *Int’l Conference on Flight Vehicles Aerothermodynamics and Re-entry Missions & Engineering (FAR)*, url: https://ora.ox.ac.uk/objects/uuid:eb4c863f-b92e-4a78-b8c6-f1e9c53df6ca

Liquid Crystal Heat Transfer Measurements in LDT

2. Donaldson, N.L., Doherty, L.J., Ireland, P.T., Merrifield, J. (2021), “Measurements of Heat Transfer on Hemispheres at Rarefied Flow Conditions using Liquid Crystals”, *8th European Conference on Space Debris*, url: https://conference.sdo.esoc.esa.int/proceedings/sdc8/paper/275

Magnetic Suspension and Balance System

3. Doherty, L.J., Donaldson, N., Owen, A. (2021), “The Oxford Magnetic Suspension and Balance System: a Brief History & Development Status”, *AIAA SciTech Forum*, doi: https://doi.org/10.2514/6.2021-1870

CXT and OPG Selected Research Papers

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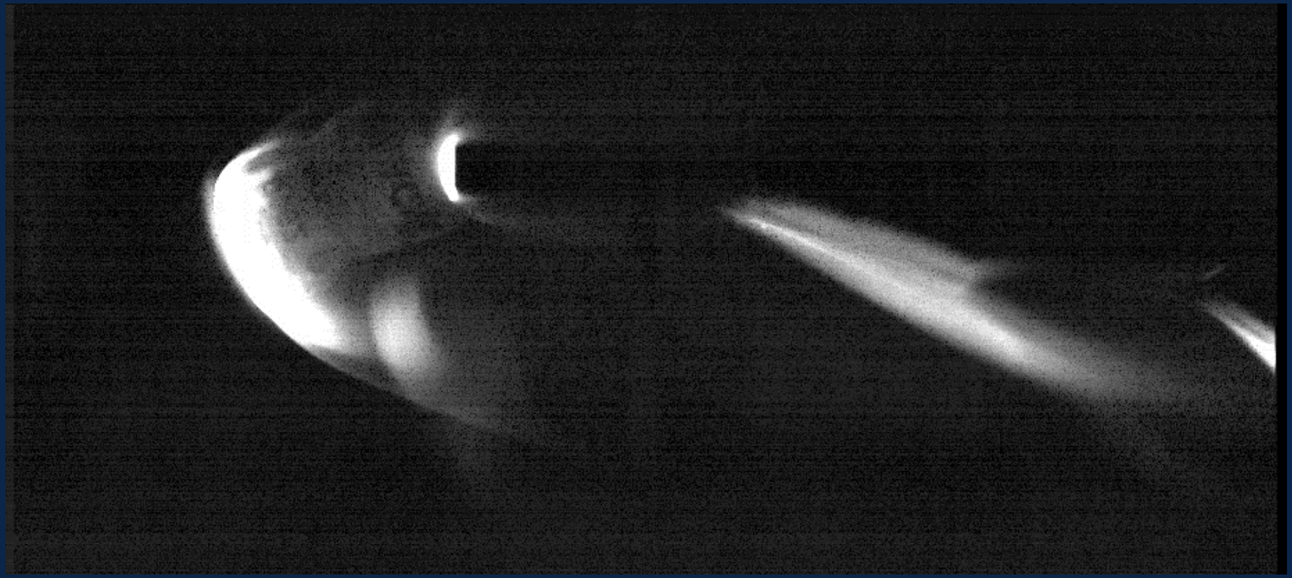
CXT Development

1. Lin, Y., Wallington, J.j>, Bui, T., van den Herik, T., James, C.M., Chang, E.W.K., Hermann, T.A. (2024), "Expansion Tube Capabilities for Studying Boost-Glide Re-entry Conditions", *3rd HiSST Conference*, Busan, Korea

OPG Development and Commissioning

2. Hermann, T.A., Chang, E.W.K., Schaefer, J., Joglekar, C., Boehrk, H. (2023), "Development of Small Scale Arc-jet Facility OPG1", *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2023-2334
3. Chang, E.W.K., Joglekar, C., and Hermann, T.A. (2023), "Integration of Arc-jet in Impulse Facility for Hypervelocity Aerothermal Testing with Ablation", *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2023-2331
4. Chang, E.W.K. and Hermann, T.A. (2023), "Commissioning Experiments for Small Scale Plasma Wind Tunnel OPG1", *34th International Symposium on Shock Waves,* Daegu, Korea
5. Schaefer, J., Chang, E.W.K., Hermann, T.A., Löhle, S. (2024), "Experimental Investigation of a Small-scale Arc-jet using Air Plasma", *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2024-2571
6. Buquet, M., Chang, E.W.K., Williams, B., Hermann, T.A. (2024), "Experimental Characterisation of a Small-scale Arc-Jet Flow using a Spatially Resolved UV-nIR Spectroscopy System", *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2024-2212
7. Valeinis, O., Chang, E.W.K., Hermann, T.A. (2024), "Development and Testing of Slug Calorimeter and Total Pressure Probes for a Miniaturized Arc-jet", *AIAA SciTech Forum*, DOI: https://doi.org/10.2514/6.2024-2886
8. Chang, E.W.K. and Hermann, T.A. (2024), "System Study of an Integrated Facility with Arc-jet and Expansion Tube for Hypervelocity Testing with Ablating Spacecraft Models", *3rd HiSST Conference*, Busan, Korea
9. Hermann, T. and Chang, E.W.K. (2024), "Optical Temperature Measuremetn in Unsteady Plasma Free Jet", *Measurement Science and Technology*, DOI: https://doi.org/10.1088/1361-6501/ad24b7



World's fastest champagne cork

Oxford T6 Stalker Tunnel, 13.2 km/s.